Introduction to Maths

Addition & Subtraction

An Ancient invented the arithmetic to the daily life things for the Goods and Services

Add * The sum of the Numbers is called Addition and (+) Symbol is used to perfor Addition

* The Addition is Commutative while the Order of the two No is Performing the result Must be the Same.

$$x + y = y$$
$$y + x = y$$

} both are Equal

* The addition is also associative like the Numbers of more than two Order are same Example $5+4+1 = 10 == 1+4+5$

Sub * The Subraction is used to Calculate the Difference between Numbers $5-3 = 2$

* The Subraction is Neither Commutative Nor Associative

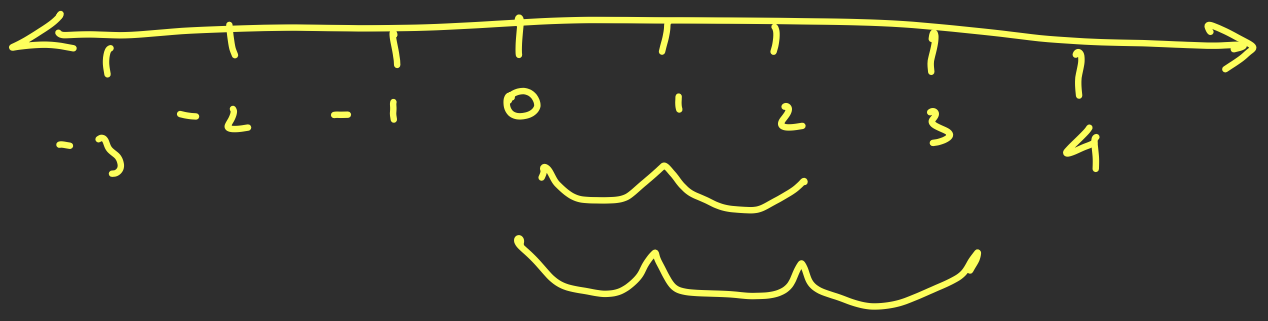
$5-4 = 1 =/= 4-5 = -1$

$5-4-1 =/= 1-4-5 = -8$

Addition And Subtraction

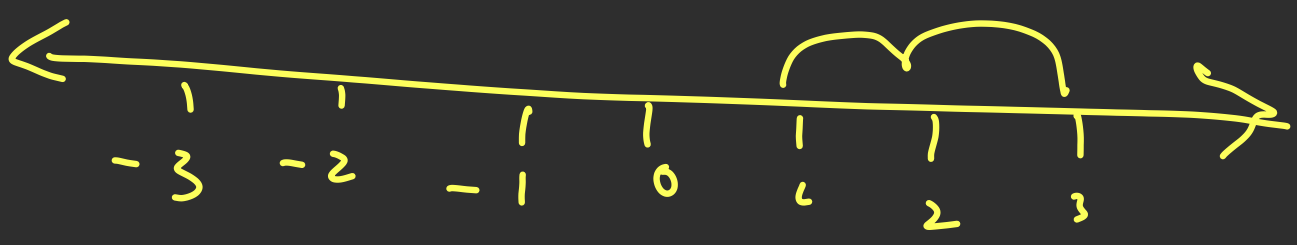
Graph Representation

$2 + 4$



2 Steps → 3 Step

5 Step → This is Add



$1 - 3 = -2$

Fraction

Fraction is a Part of the whole Number.

$\frac{1}{2}$ → 1 → Numerator
2 → Denominator

No of whole part

No of Part Separated

Types of Fraction

Proper Fraction = $\frac{1}{2}$ The Value become less than (1) so.

Improper Fraction = $\frac{5}{5}$ The Value become Greater Then (1)

Mixed Fraction is Used to Repeat the Improper Fraction in Good Way

Comprehension

Improper to Mixed

$\frac{17}{7} = 2\frac{3}{7}$

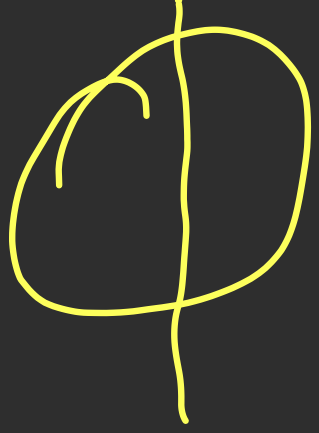
$\frac{13}{3} = 4\frac{1}{3}$

$\frac{41}{5} = 8\frac{1}{5}$

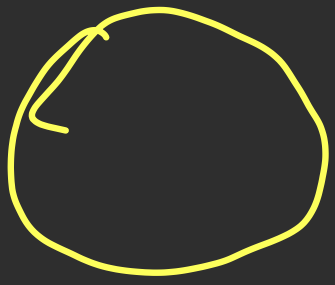
$4\frac{8}{3}$

$\frac{12+8}{3}$

$\frac{20}{3}$



$= \frac{1}{2}$



$\frac{1 \cdot 5}{2}$

$\frac{1}{\cancel{6}\ 1}$

$\cancel{5}$

$1\frac{1}{6}$

$\frac{6}{\cancel{5}}$

Fraction

A Part of whole

① Whole 1 whole 2 parts

$\frac{1}{2}$

② whole ③ Parts

$\frac{2}{3}$

In whole ③ we need to Give 3 equal Parts to each

Totally we have 5 apples and we have only (2) Persons

$\frac{5}{2}$ balance (1) $\frac{1}{2}$

$2\frac{1}{2}$ Mixed Fraction

And It is Improper Fraction

We can say $\frac{13}{5}$ we have $\frac{1}{5}$ units 13 times instead of say $\frac{3}{5}$ as (3) out of (5) we can say 3 units and each unit is $\frac{1}{5}$ so if its reached more than (1) this sentence make more good and proper most invented improper fraction for calculation else we

Used the Mixed Fraction
For the Calculation

$2\frac{3}{3} + 4\frac{5}{4}$ ✗

$\frac{9}{3} + \frac{21}{4}$ ✓

Thus Come able to
Calculate more Broad

The Denominator is Smaller
than Numerators Because of
Improper Fraction so
Why Is Improper.

Place Values.

We have only

1 2 3 4

5 6 7 8

9

0

This was invented by

Hindus became we

Evolved with 10

Fingers 10

1 2 3 4 5 6 7 8 9 0

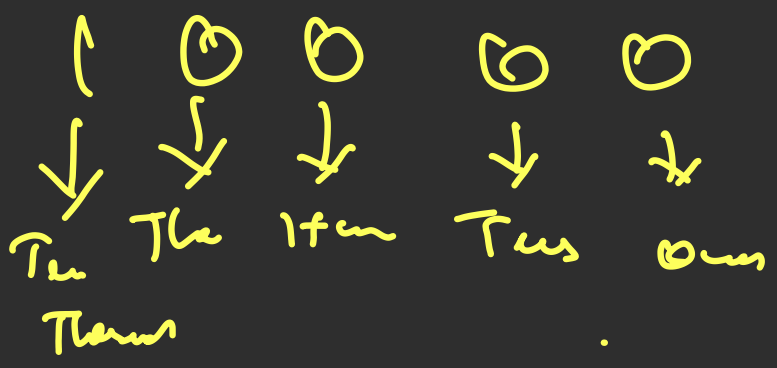
(9) Unit Place

After (9) what to do we have Confusion so we introduced Place Value

10 After
↓ one
Tenth Place

100
Hundredth

Power of 10 We Calculate the Place Values



Decimals

Decimal is Used to Create more Complex Calculation For the Fraction .

$6 + \frac{2}{3}$ Instead $6 + 0.12$

Like this

Decimal

k/y Existe → Actually the Fraction solved the mechanism of Parts of whole Number but in the Advanced Mathematical Calculation The Fraction become worse. so Here is the Mechanism

k/e Already Know Place Value

4 3 2 1
Ten H. ten Unit

Every Left Side is 10x bigger

→ Every Right Side is 10 ÷ which is Smaller .

100 10 1 $\frac{1}{10}$ $\frac{1}{100}$ $\frac{1}{1000}$

0.1 is $\frac{1}{10}$

0.2 0.3 ... 0.9

1.0 = 1

$\frac{1}{100}$ 0.01 0.02 ... 0.09

0.10 ... 0.99

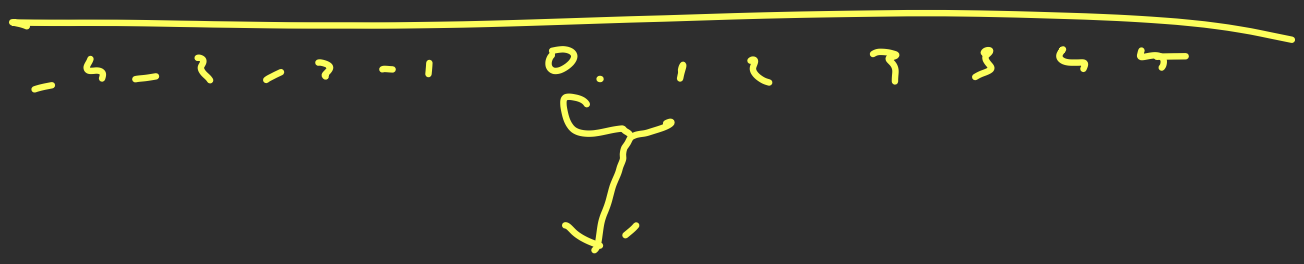
1.00 so it takes

100 of $\frac{1}{100}$ is 0.01

$\frac{x+9}{10}^2$

$\frac{19}{10}$

$\frac{10}{10}$



Percentages

For Every

Per - 1

Cent = 100

Every 100 we have

Quantity

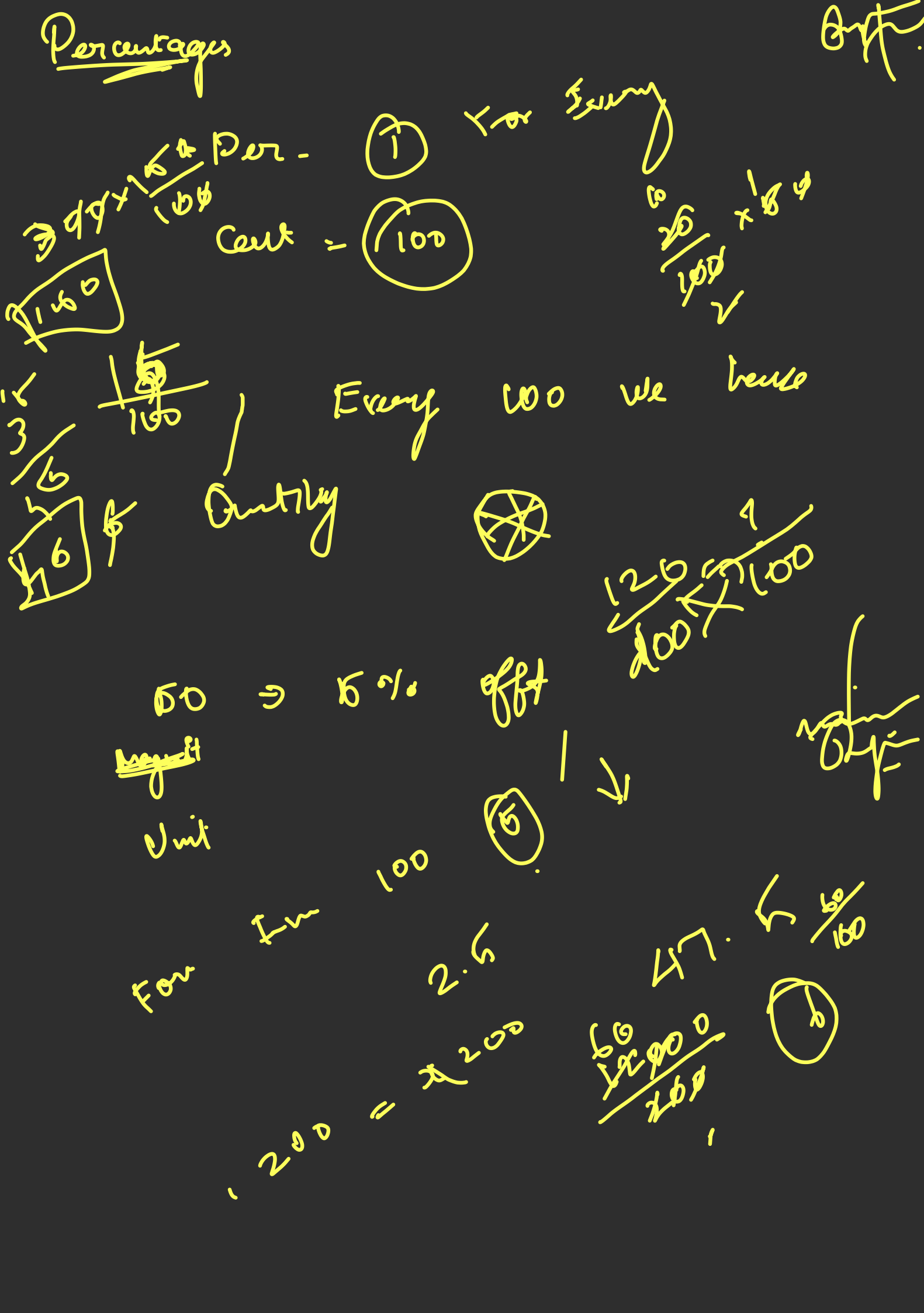
Profit

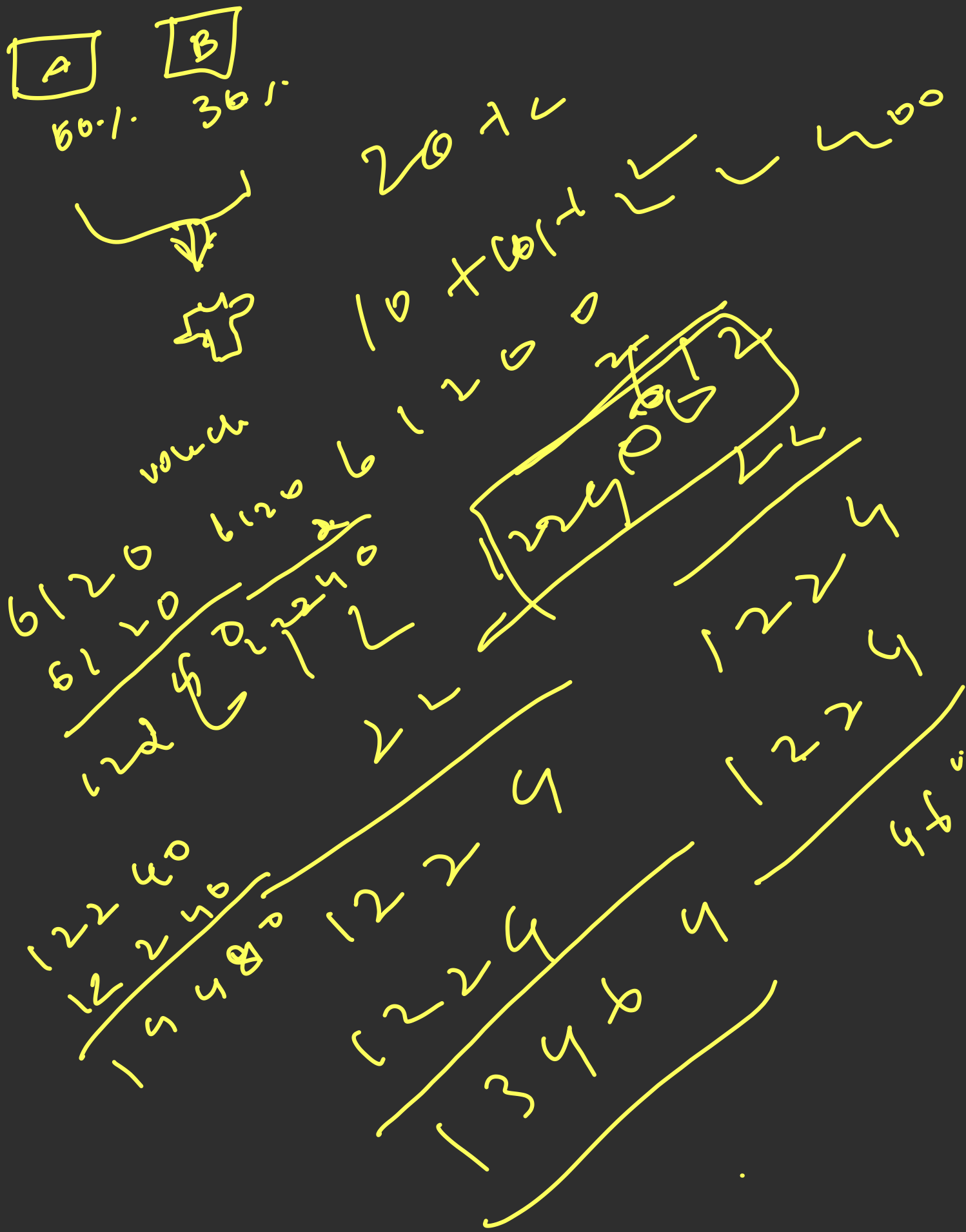
Unit

For Rs 100

2.5

47.5




A
B
60%
36%

Exponent

$e^x \Rightarrow$ Stacking

↓ base

$5^2 = 5 \times 5 \Rightarrow 25$

$5^3 = 5 \times 5 \times 5 = 125$

Expressions

$x^a * x^b = x^{a+b}$

Example

$2^3 \times 2^5 = \frac{2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2}{\text{Total of } 2s}$

2^{3+5} ! We can Add

For Addition we can Add both the Exponent if Have the same base.

$$\frac{x^a}{x^b} = x^{a-b}$$

$x^a \cdot x^b$

$2^3 \cdot 2^4 = (\text{same base}) \; BODMAS$

$$\frac{2^5}{2^3} = \frac{2 \times 2 \times 2 \times 2 \times 2}{2 \times 2 \times 2} = \frac{1}{2}$$

$$\frac{2}{2} = 1$$

10 ve Cash Jaise

Sol

$2 \times 2 = 2^2 = 4$

$$\frac{x^a}{x^b} = = x^{(a-b)}$$

Hence Proved

What if Exponent has negative value

$$\frac{2^3}{2^5} = \frac{2 \times 2 \times 2}{2 \times 2 \times 2 \times 2 \times 2} = \frac{1}{2^2}$$

Hence

$2^4 = 2 \times 2 \times 2 \times 2$

$2^3 = 2 \times 2 \times 2$

$2^2 = 2 \times 2$

$2^1 = 2$

$2^0 = \frac{2}{2} = 1$

$2^{-1} = \frac{1}{2}$

$2^{-2} = \frac{\frac{1}{2}}{2}$

$2^{-3} = \frac{1}{8}$

This is the Pattern

So Anyway the Negative Exponent is

$2^{-3} = \frac{1}{2^3}$

2

$((x)^a)^b$

Will be

$x^{a \times b}$

$(2^3)^2$

$(2 \times 2 \times 2)^2$

$(2 \times 2 \times 2) \times (2 \times 2 \times 2)$

2^{3n}

2^6

Prime Numbers

* The Prime Number is the Atoms of the big Number

* When ever you Take Any Number that's Fifth Greater by the Prime Number else No Number has be Splitted

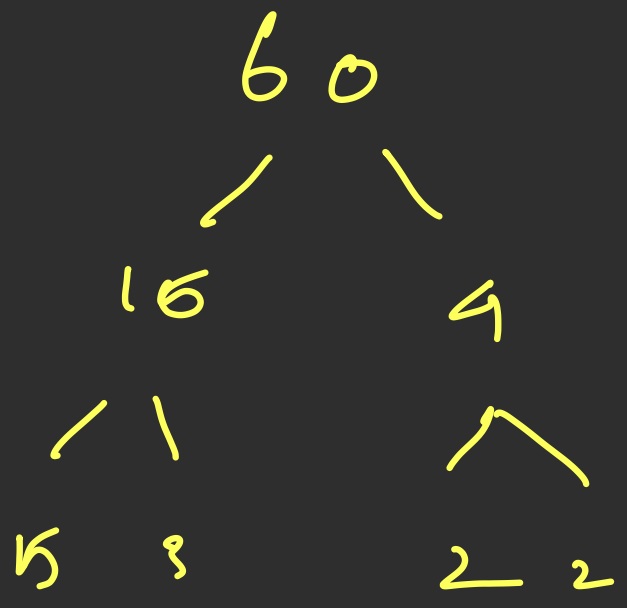
* We dont Know How the Prime Numbers are Made

Prime Factorization

We Need to Split the Number by its Factors Until We Ended Up with Prime Numbers

Examples

We Take 60



$2^2 \cdot 3 \cdot 5$ This is the Prime Factorization

LCM

Least Common Multiple

52 → 26, 2 → 13, 2

$2^2 \cdot 13$

72 → 9, 8 → 3, 3, 4, 3 → 2, 2

$2^2 \cdot 3^2 \cdot 4$

$2 \cdot 2 \cdot 13$

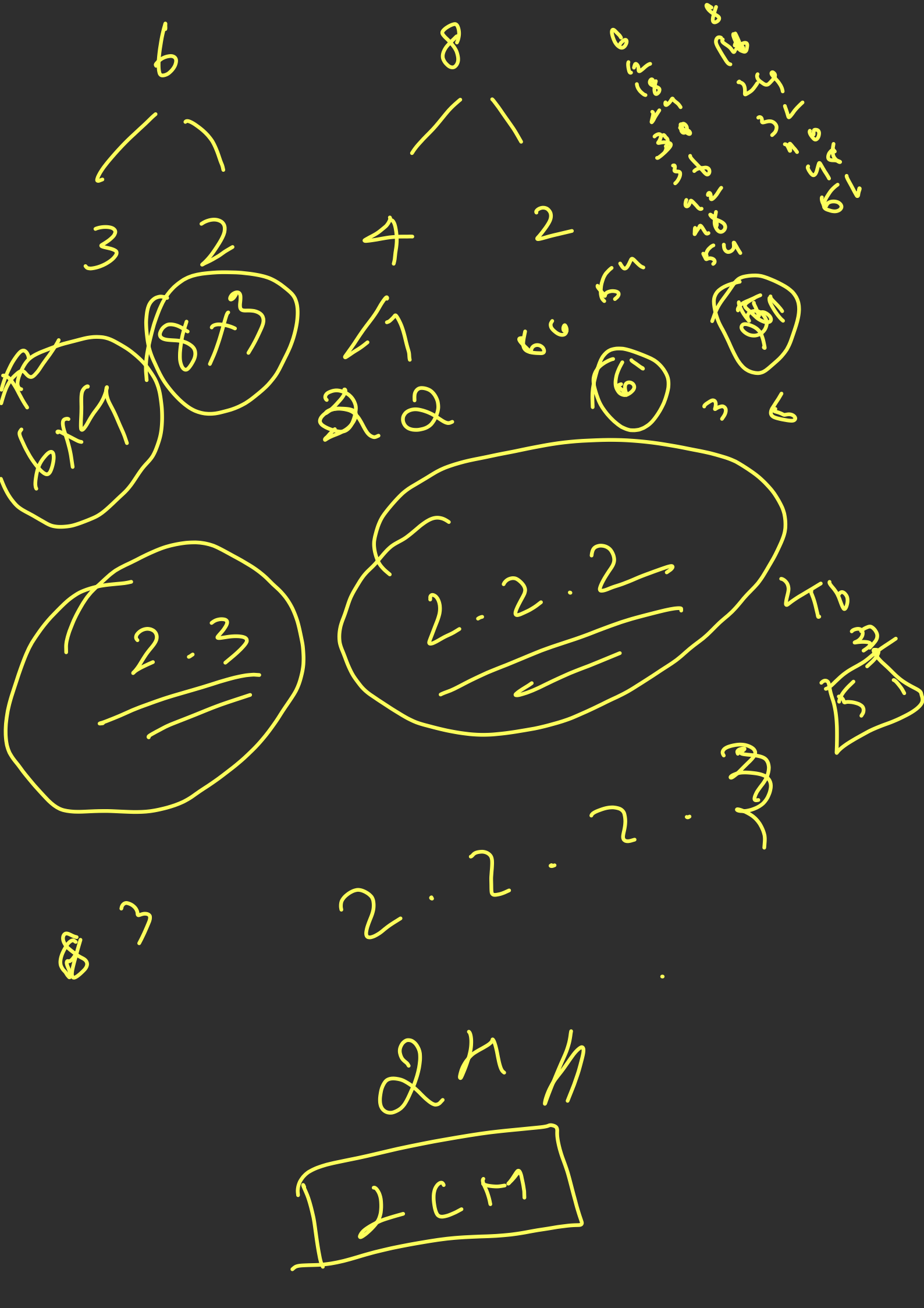
$1 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3 \cdot 4$

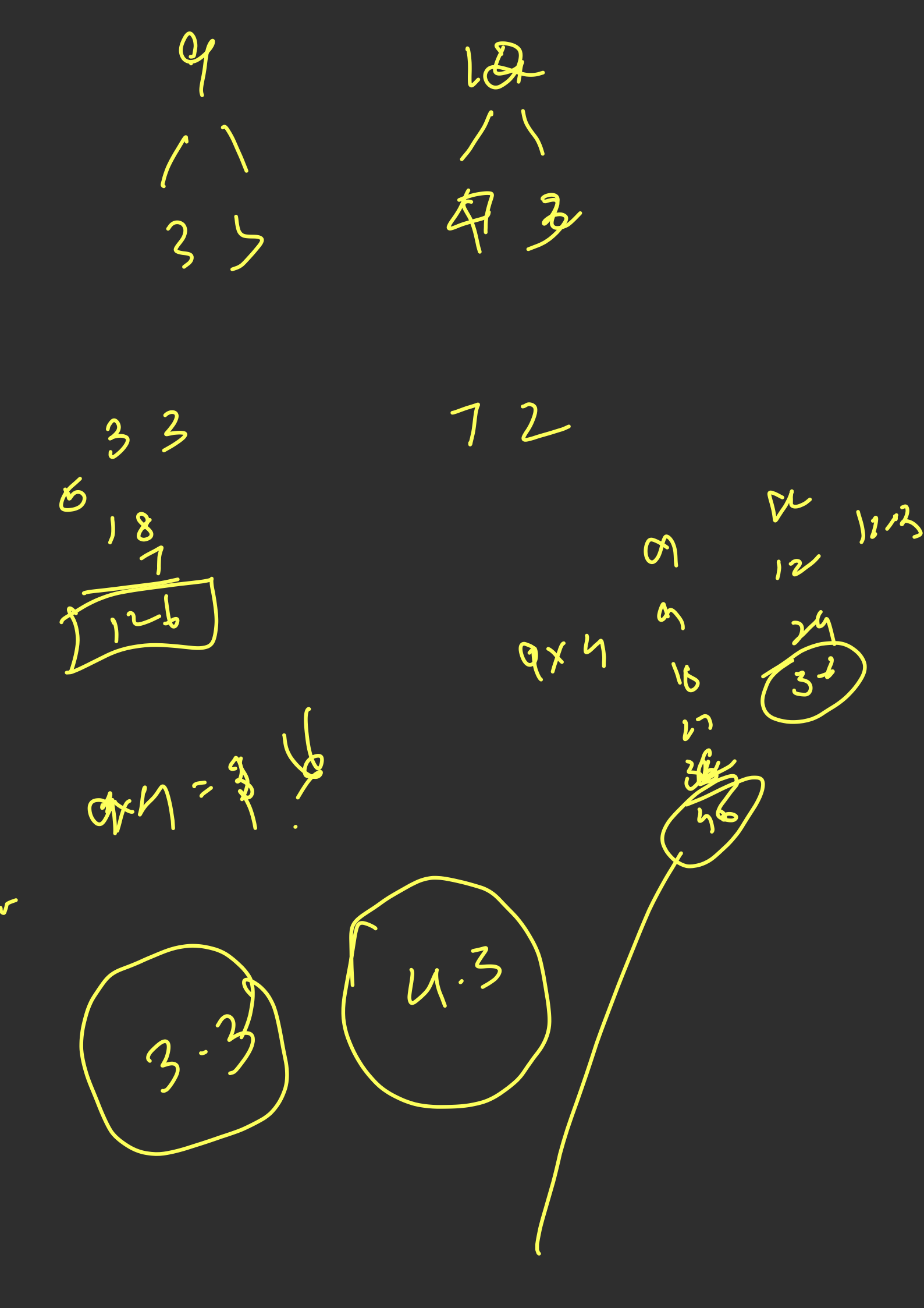
4

1
5, 3
1
5, 3
3
1
5, 3
4

4
5, 2
2

$2^2 \cdot 3^2 \cdot 4$


6
8
3
2
4
2
2.3
2.2.2
2.2.2.3
24
LCM



Greatest Common Factor

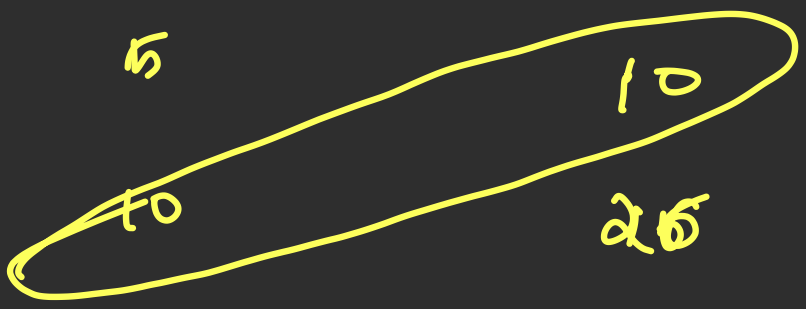
5 10

1 . 1

5 10

10 20

15 30

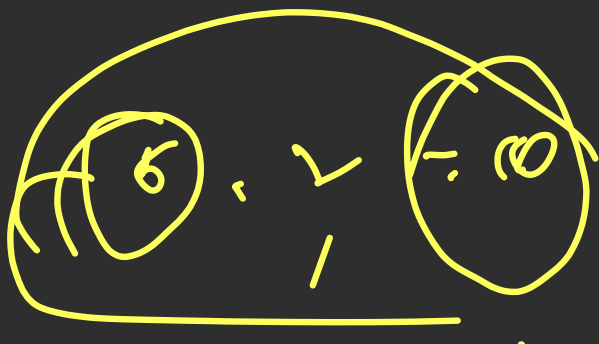


This is the Greatest Common Factor

5 10

5 / 2 5

5 . 2 . 5



10 20

5 2 10

5 2

5 . 2 2 . 2 . 5

$2^2 . 5 = 20$

Least Common Multiple

5 → 1, 5, 10, 15, 20, 25

10 → 10, 20, 30, 40, 50

5 d.5 25 '0

Greatest Common Factor

5 → 1, 5

10 → 1, 2, 5, 10

5 10

1 1

2 5

= 5

28

28

1

Least Common Factor

28 → 28

56

28

1 1

14 2

28 2

1 1

7 2

14 2

1

2

2 . 2 . 7

2 . 2 . 2 . 7

LCM

Least Common Multiple

8

8

16

24

32

40

48

56

56

12

12

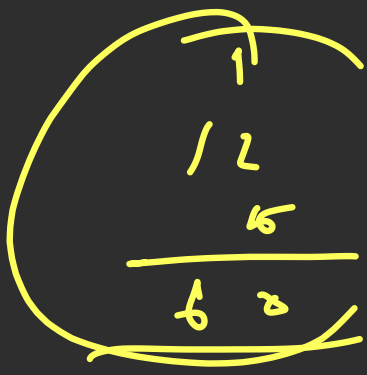
24

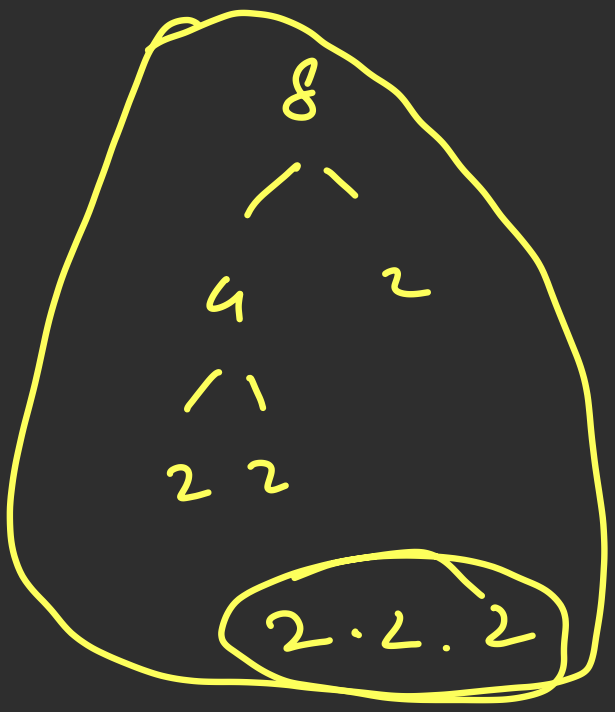
36

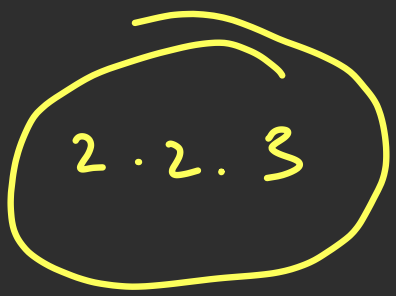
48

60

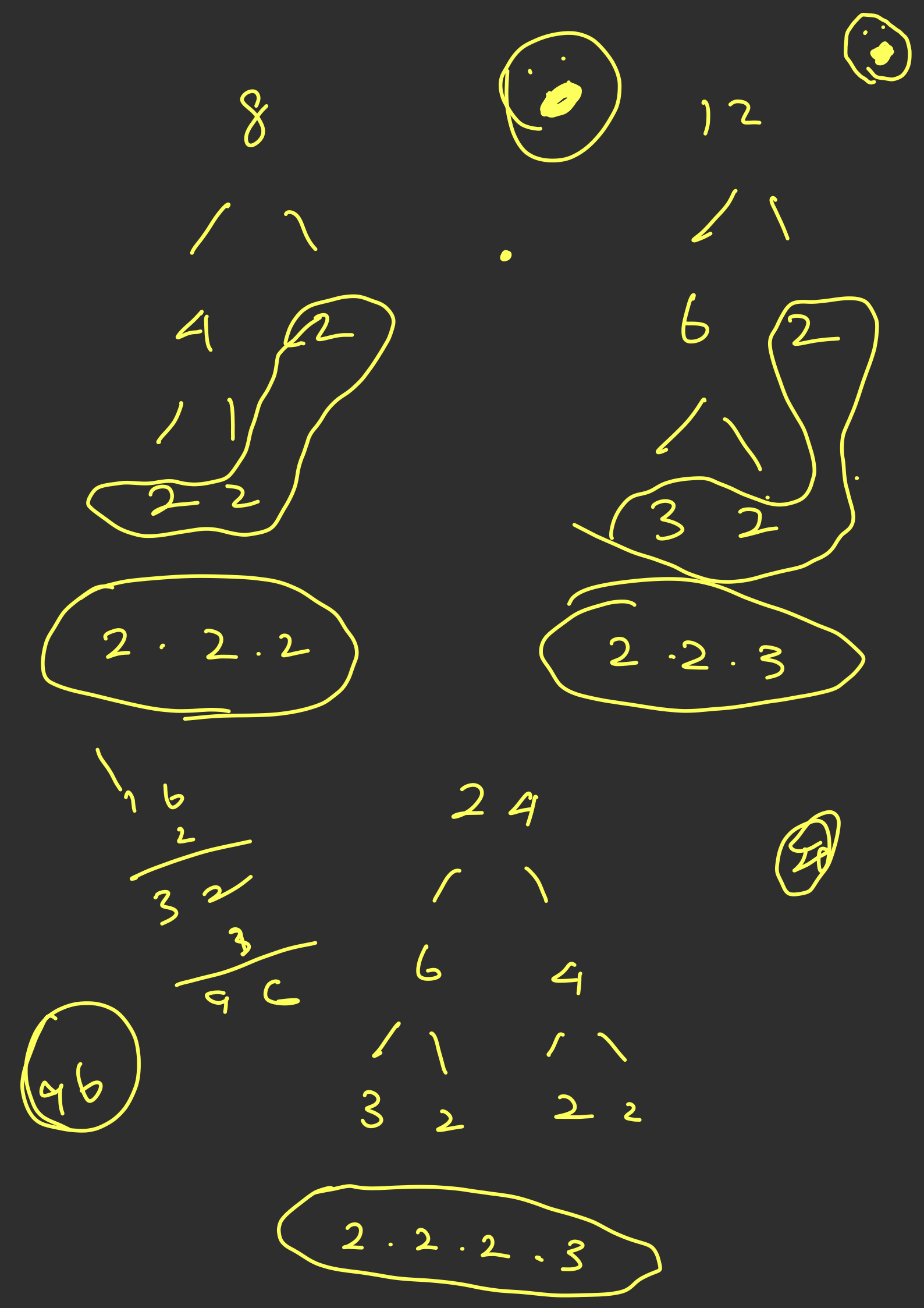
12
/ 4
6 2
/ \
3 2







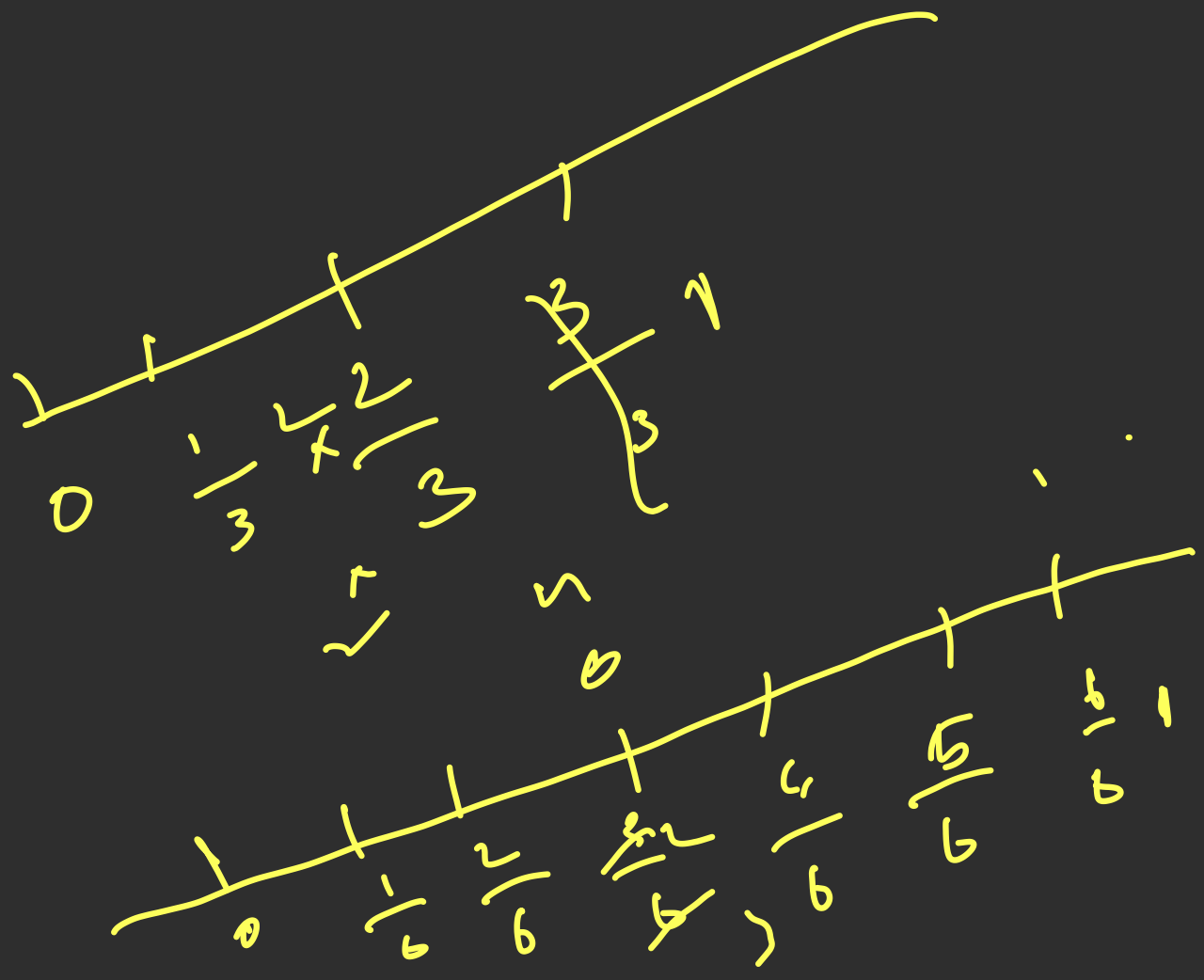
$2 \cdot 2 \cdot 2 \cdot 3 = 24$


8
4 2
2 2
2 · 2 · 2
12
6 2
3 2
2 · 2 · 3
16
2
3 2
3
8 8
48
24
6 4
3 2
2 2
2 · 2 · 2 · 3

Addition & Subtraction of Fract...

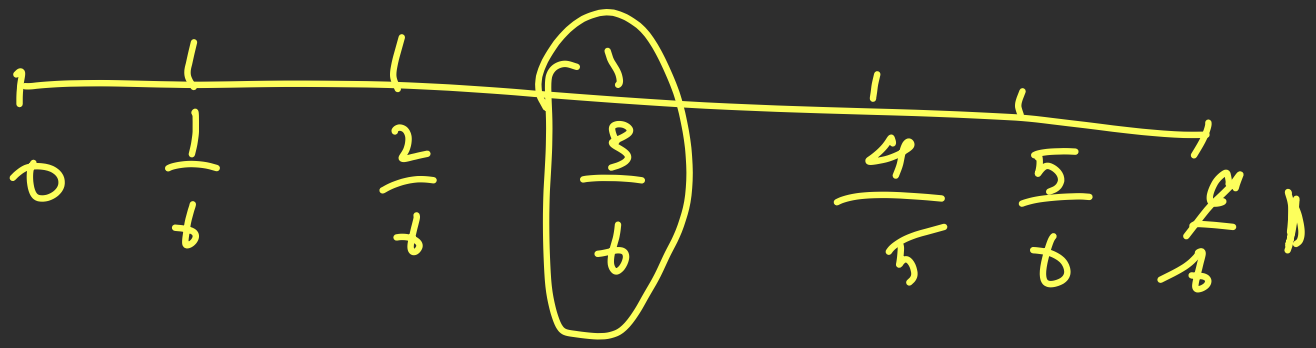
$$\frac{2}{6} + \frac{4 \times 2}{3 \times 2}$$

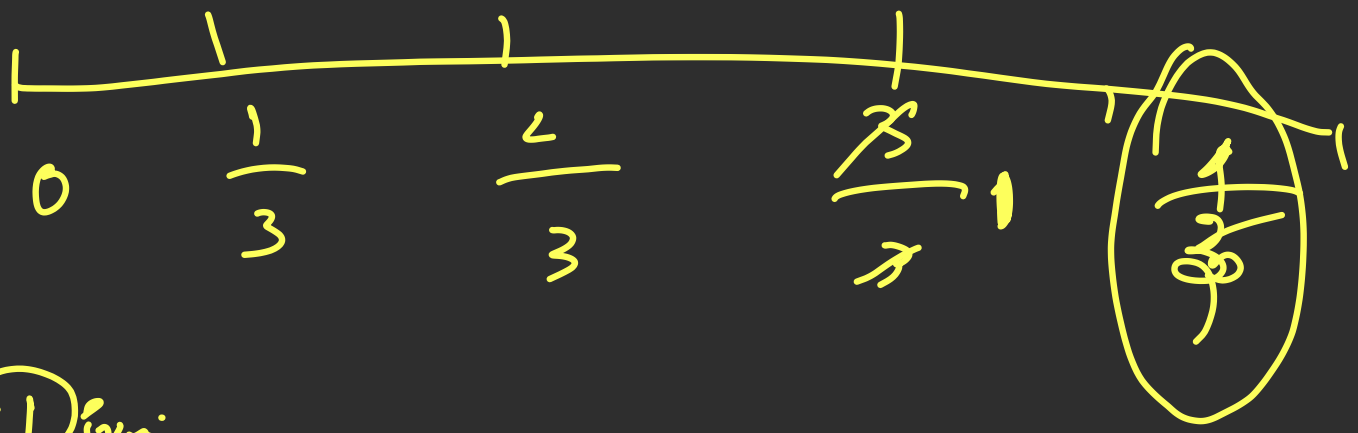
$$\frac{8}{6} \qquad \frac{4}{3}$$



Multiplication & Division

$$\frac{2}{6} \times \frac{4}{3}$$

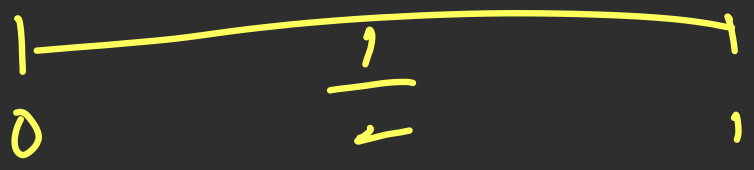




Divi

$$\frac{2}{4} \div 2$$

$\frac{1}{2}$ $\frac{2}{4}$ $\frac{3}{4}$



$1 \div \frac{1}{2} = 2$

$\frac{1}{2}$

$\frac{3}{4}$

$1 \div \frac{1}{2} = 2$

$\frac{1}{2} \times x = 1$

$x = \frac{1}{\frac{1}{2}}$

$\frac{1}{\frac{2}{1}}$

$1 \times \frac{4}{3} = \frac{4}{3}$

$\frac{3}{4} \times x = 1$

$x = 1 \times \frac{4}{3}$

$x = \frac{4}{3}$

Average

For Logic Calculation

we have Number of Set

we Need Some Calculation

for This Time

For Example you have Some

Set Laps 100, 120, 220, 220, 180.

Average Km for The Laps

we Need to Calculate

$$\frac{\text{Sum of Num}}{\text{No. of Item}}$$

$$\frac{100 + 120 + 220 + 220 + 180}{5}$$

$$= \frac{840}{5} = 168 \text{ Km}$$

$$\begin{array}{r} 168 \\ 5\overline{)840} \\ 5 \\ \hline 34 \\ 30 \\ \hline 40 \\ 40 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 100 \\ 120 \\ 220 \\ 220 \\ 180 \\ \hline 840 \end{array}$$

Average Speed

There is the Difference of Value

Range = Least & Highest Value

Highest - Least

$$220 - 100 = 120$$

Mean = Average =

$$= \frac{840}{5}$$

$$= 168$$

Median

Acending Order and Make It Middle Value

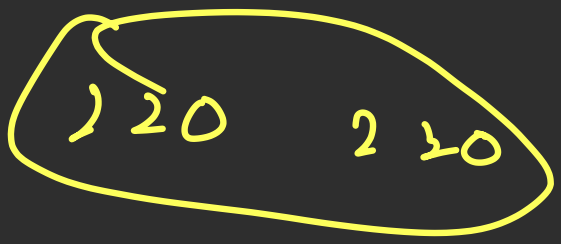
100 120 180 220 220

↓

180 is Median

Mode - The Number Which Repeats Frequently

100 120 180 120 220



This is

Algebra → Finding the Culprits using Variables

Associative = $a+b = b+a$ ⇒ works on Add & Multi

Commutative = $a+(b+c) = (a+b)+c$ ⇒ works Add & Mul

(*) Distributive = $a(b+c) = (a+b)c$

$3x^2+6x = 3x(x+2)$

Additive Identity

$a+0 = 0$

Anything + 0 is

Same result

Multiplicative Identity

$a \cdot 1 = a$

Except '0'

Inverse Property of addition

$x + (-x) = 0$

Inverse Property of Same

$x \cdot \frac{1}{x} = 1$

1

Algebraic Equation

$x + 6 = 13 \Rightarrow x + 6 - 6 = 13 - 6$ $\boxed{x = 7}$

$x - 4 = 1 \Rightarrow x - 4 + 4 = 1 + 4$ $\boxed{x = 5}$

$5x = 70 \Rightarrow \frac{5x}{5} = \frac{70}{5} = \boxed{14}$

$\frac{x}{3} = 9 \Rightarrow \frac{x}{3} \times 3 = 9 \times 3$ $\boxed{27}$

$3x - 2 = 5x + 5 \Rightarrow -2 = 2x + 8 \Rightarrow 2x = -10 \Rightarrow -\frac{10}{2}$

$\boxed{x = -5}$

$2(x + 6) = 4(x + 1) \Rightarrow 2x + 12 = 4x + 4$

$\Rightarrow 12 = 2x + 4 \Rightarrow 2x = 12 - 4 = 2x = 8$ $\boxed{x = 4}$

$2(x + 6) = 4(x + 1) = \frac{4(x+1)}{4}$

$x + 6 = 2x + 2$

$6 = x + 2$ $\boxed{x = 4}$

$\frac{(5x - 4)}{3} = 2(x + 4) = 5x - 4 = 6(x + 2)$

$5x - 4 = 6x + 12$

$x = -12 - 4$

$\boxed{x = -16}$

Sally = $2x - 3$ = $24 - 3 = 21$

Crabby = $x - 12$

$x + y + z = 36$

$x + x+3 + x+6 = 36$

$3x + 9 = 36$

$3x = 27$

$x = \frac{27}{3} = x = 9$

$x = y + 6$

$2y + 3x = 68$

$2y + 3(y+6) = 68$

$2y + 3y + 12 = 68$

$5y + 12 = 68$

$y = \frac{80}{5}^{16}$

$y = 16$

$x = 22$

$2y + 3y + 18 = 68$

$5y = 168 - 18$

$y = 10$

$x = 16$

$x + y = 70$

$y = 2x - 5$

$x + 2x - 5 = 70$

$3x - 5 = 70$

$3x = 65$

$x = \frac{65}{3}^{2}$

$x = 22$

$\frac{2}{70}^{10}$

$\frac{25}{45}$

45

$\frac{2}{15}$

$\frac{15}{3}$

Inequalities in Algebra

Rather than equal like $x = 5$

The Inequalities Refers $x < 5$

That could be Any Number

Some Calculation we can do

$x - 4 > 3 \Rightarrow x > 7$

$3x + 2 < 20 \Rightarrow 3x < 18 \Rightarrow x < 6$

$3 - x > 17 \Rightarrow -x > 20 \Rightarrow x < -20$

$2x + 9 < 4x + 3 \Rightarrow 2x + 6 < 4x$

$1 < x$ $\boxed{x < 1}$ $2 < 4x - 2x$

$2x + 9 < 4x + 3 \Rightarrow 2x + 6 < 4x$

$6 < 4x - 2x$

$6 < 2x$

$3 < x$ $\boxed{x > 3}$

Square root and Other roots

$\sqrt{9} = 3$

$\sqrt[3]{27} = 3$

$\sqrt{}$ → Radical Symbol

Exponent Inverse

$\sqrt{x} \Leftarrow x^2$ $\quad 3^2 = 9 \quad \sqrt{9} = 3$

$\sqrt{x} = x^{\frac{1}{2}}$ because $\sqrt{x} \times \sqrt{x} = x$

$x^{\frac{1}{2}} \times x^{\frac{1}{2}} = x$

Cube roots

$\sqrt[3]{x} \Leftarrow x^3$ $\quad \sqrt[3]{x} = x$

$x^{\frac{a}{b}} = \sqrt[b]{x^a}$

$x^{\frac{1}{2}} = \sqrt[2]{x}$

$x^{\frac{1}{3}} = \sqrt[3]{x}$

$\frac{x^3}{x^1} \quad \frac{x^2}{x} \quad \frac{x^1}{x^1} \quad \frac{x^0}{x^1} \quad \frac{x^{-1}}{x}$

1) $(2x^2y^4z^6)^3$

$8x^6y^{12}z^{18}$

$8x^6y^{12}z^{18}$

2) $\frac{x^{(2y+1)}}{x^{(2y-1)}}$

$x^{[(2y+1)-(2y-1)}$

$x^{[2y+1-2y+1}$

x^2

3) $\left(\frac{xy^2}{x^2y}\right)^2$

$\left(\frac{x^2y}{xy^2}\right)^2$

$\frac{x^4y^2x^1}{x^2y^4z^2}$ $\frac{x^2y}{xyz}$ $\left(\frac{x}{y}\right)$

$\frac{x^2y^1}{xy^2}$

$\left(\frac{x}{y}\right)^3$

4) $\frac{\left(x^{\frac{1}{3}} \cdot x^{\frac{2}{3}}\right)^2}{x^{3/2}}$

$\frac{x^{\frac{2}{3}} \times x^{\frac{4}{3}}}{x^{\frac{3}{2}}}$

$x^{\frac{2}{3}+\frac{4}{3}=\frac{6}{3}=2}$

$\frac{x^2}{x^{\frac{3}{2}}}$ $x^{\frac{4}{2}-\frac{3}{2}}$

$\frac{4-3}{2}$

$\frac{1}{2}$

$x^{\frac{1}{2}}$

$\left(\frac{xy^2}{x^2y}\right)^{-2}$ $\left(\frac{x^2y}{xy^2}\right)^2$ $\frac{x^4y^2}{x^2y^4z^2}$

$\sqrt{x+9} = 5$

$\sqrt{x} + \sqrt{9} = 5$

$\sqrt{x} + 3 = 5$

$\sqrt{x} = 5 - 3$

$\sqrt{x} = 2$

$\boxed{x = 4}$

$\sqrt{x+y} = 10$

$\sqrt{16+9} =$

$\sqrt{25} = 5$

$\sqrt{16} + \sqrt{9} =$

$4 + 3 = 7$

$\sqrt{x+9} = 5$

$(x+9)^2 = 5^2$

$x + 9 = 25$

$x = 16$

$x^2 + 11 = 36$

$x^2 = 36$

$x^2 = 25$

$\boxed{x = \pm 5}$

$x^2 - 9 = -17$

$x^2 = -26$

$x = \pm\sqrt{-8}$

$x = \sqrt{-8}$

$x = \sqrt{4 \times 2}$

$x = 2\sqrt{2}$

$x = -8$

$x = -\sqrt{4 \times 2}$

$x = -2\sqrt{2}$

$\boxed{x = \pm 2\sqrt{2}}$

$\sqrt[3]{x^2+15} = 4$

$x^2 + 15 = 4^3$

$x^2 = 64 - 15$

$x^2 = 49$

$x = \sqrt{49}$

$\boxed{x = 7}$

$\sqrt{x^3+9} = 6$

$\sqrt{x^3+9} = 6$

$x^3 + 9 = 36$

$x^3 = 27$

$\boxed{x = 3}$

$\begin{array}{r|l} 2 & 46 \\ \hline 4 & \\ \hline & 64 \end{array}$

$\begin{array}{r|l} 2 & 36 \\ \hline 9 & \\ \hline & 27 \end{array}$

Introducing to Polynomials

$2x$ → monomial

$2x - 1$ → binomial

$2x^2 - 2x - 1$ → Trinomial

$2x^3 - 2x^2 - 2x - 1$ → Polynomial

After all

It is an kind of an Algebraic equation

$$2x^{3} - 2x^{2} - 2x - 1$$

The Highest Power we Call that as degree

and we Call the

If the degree 2 → Quadratic equation. If '3' → Cubic Polynomial Quantic for 4^{th} and Above

There is Some Structure

$$a_n x^n + a_{(n-1)} x^{n-1} + a_{(n-2)} x^{n-2} \cdots + a_0$$

Subscript in Daily

It's refer the Order of The Sequence

$P_1 \quad P_2 \quad P_3$

Thus Subscript Refer Order.

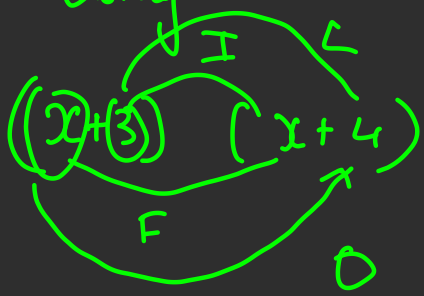
Addition & Subtraction In Polynomial

$$3x^3 + 5x^2 - 2x + 9$$
$$x^3 - 4x^2 + 7x - 11$$
$$4x^3 + x^2 + 5x - 2$$

$$x^3 + 6x^2 + 6x - 1$$
$$-5x^3 + 5x^2 + x - 8$$
$$2x^3 + 5x^2 + 7x - 9$$

Multiplying Binomial or Trinomial

Using FOIL Method First Outer Inner Last



$(x+3)(x+4) \Rightarrow x^2 + 4x + 3x + 12$

$\Rightarrow x^2 + 7x + 12$

$(x-2)(x^2+1) \Rightarrow x^3 + x - 2x^2 - 2$

$\Rightarrow x^3 - 2x^2 + x - 2$ //

$(2x-9)(3x+5) \Rightarrow 6x^2 + 10x - 27x + 45$

$\Rightarrow 6x^2 - 17x - 45$ //

$(2+3) \times (3+4)$

$6 + 8 \times 9 + 12$

$14 \neq 21$

$\begin{array}{r} 21 \\ 14 \\ \hline 35 \end{array}$

$5 \times 7 = 35$

Factor Polynomial

$(x+2)(x-3) = x^2 - x - 6$

FOIL ⟶ Quadratic
⟵

Quadratic Polynomial Equation

$ax^2 + bx + c = 0$ → Structure

As we said before Quadratic to

$(x+m)(x+n)$

$x^2 + xn + xm + mn$

$x^2 + (m+n)x + mn$

So If we have in this Structure which is mn multiplied by 1100 Number will also Get added by other 1100 Number

$x^2 + 8x + 7 = 0$

$(x+7)(x+1) = 0$

7 × 1 = 8

$x = -7, x = -1$

We Got 1100 Solution

$2x^2 - 4x + 2$

$(2x+m)(x+n)$

$*2*2$
m
$-2 -2$
m
$+2 -2$
$-2 +2$

$2x^2 + 2xn + mx + mn$

$1 \quad 2$

$2x^2 + 4x + x + 2$

$2x^2 + 5x + 2$

$2x^2 + 2x2 + -2x + 2 \cdot 2$

$2x^2 + 4x - 2x + 0$

$2x^2 + 2x$

$3x^2 + 5x - 2$

$ax^2 + bx + c$

$(3x + m)(x + n)$ binomial Produ

$x^2 + mx + nx + mn$

$x^2 + x(m+n) + mn$

$ax^2 + bx + c$

$(3x + m)(x + n)$:

$3x^2 + mx + 3xn + mn$

$3x^2 + mx + 3xn + mn$

$3x^2 + x(m + 2(n) + mn$ — (1)

$3x^2 + x(2(m + n) + mn$ — (2)

It Can Produce two equation

$(3x+m)(x+n)$

$3x^2 + 3xn + mx + mn$

* Something the two Number Multiplied will Give mn

$3x^2 + x(3n+m) + mn$

$(3x+2)(x+1)$

$3x^2 + 3x + 2x + 2$

$3x^2 + 5x + 2$

$(3x-2)(x+1)$

$3x^2 + 3x - 2x - 2$

$3x^2 + x - 2$

$(3x-1)(x+2)$

$3x^2 + 6x - x - 2$

$3x^2 + 5x - 2$

Quadratic Polynomial by Factoring

1.) $x^2 - x - 20$

$-1, -20 \rightarrow -5 \times 4$

$x = +5 \quad x = 4$

$\boxed{x = 5, \; x = -4}$

2.) $x^2 + 8x + 7 = 0$

$x = -7, \; x = -1$

7×1

$(-1) \quad 1 - 8 + 7$

$8 - 8 = 0$

3) $2x^2 - 4x + 2 = 0$

$(x-1)(x-1)$

$x^2 + mx + xn + mn$

$x^2 + x(m+n) + mn$

$ax^2 + mx + anx + mn$

$ax^2 + x(m + an) + mn$

$0 \qquad x(am + n)$

$2x - 2 = 0$

$2x = 2$

$2x - x = 2 - 1$

$\boxed{x = 1}$

$2x^2 - 2x - 2x + 2$

$2x^2 - 4x + 2 \qquad 2(-1) + 2$

$(2x-2)(x-1) \qquad -2 - 2 \qquad -4$

$(2x-1)(x-2)$

$ax^2 + bx + c = 0$

multiply b/w $(x+m)(x+n)$

$x^2 + mx + nx + mn$

2×1

-2×-1

$2x - 2 - 1$

$-4 - 1 \qquad (x+m)x \cdot n$

$-5 \qquad (2x-2)(x-1)$

$2 * -1 \quad x = 2$

$-2 + 2$

$-2 - 1 \qquad -2 - 1$

$-4 \qquad 2^{m \times n} + m \times n$

$2(-1) + -2$

$2(-1) - 2 \qquad 2 \times 1$

$-2 - 2 \qquad -4x$

$2x^2 - x + 4x^2 \quad x^2$

$2x^2 - 3x + -$

Quadratic equation Solving by Completing Square

Let's Examine the Square roots

$x^2 + 2x + 1$ → by Factoring we either Have $+1, +1$ → which give (2)
$-1, -1$

$x^2 + 2x + 1 = (x+1)(x+1) = 0$

$(x+1)^2 = 0$ since (-1) can only be the solution to Get the equation '0'

Some equation Can't by Solved by the Factor So why we Have Completing Square

$x^2 + 2x - 6 = 0$

Since No Factor 3 2, 2 3, 1 6, 6 1 Can Solve This

Soc

$x^2 + 2x = 6$

If we Need the same Factors we Need to Divide the 'b' term by 2 $\frac{2}{2} = 1$ and Square it

$x^2 + 2x + 1 = 6 + 1$

$(x+1)(x+1) = 6$

$(x+1)^2 = 6$

$\sqrt{(x+1)^2} = \sqrt{6^2}$

$x+1 = \sqrt{6}$

$x = \sqrt{6} - 1$

This is the Answer.

2) $x^2 + 6x - 2 = 0$

Since It does have Factors

$x^2 + 6x = 2$

Now dividing by '2' the b term & square it

$\frac{6}{2}3 = (3)^2 = 9$

$x^2 + 6x + 9 = 2 + 9$

$(x+3)^2 = 11$

$\sqrt{(x+3)^2} = \pm\sqrt{11}$

$x = \pm\sqrt{11} - 3$ //

3x

3) $3x^2 + 15x - 8 = 0$

$\frac{3x^2}{3} + \frac{15x}{3} = \frac{8}{3}$

$x^2 + 5x = \frac{8}{3}$

$x^2 + 5x + \left(\frac{5}{2}\right)^2 = \frac{8}{3}$

$x^2 + 5x + \frac{25}{4} = \frac{8}{3} + \frac{25}{4}$

$x^2 + 5x + \frac{25}{4} = \frac{32 + 75}{12}$

$x^2 + 5x + \frac{25}{4} = \frac{107}{12}$

$\left(x + \frac{5}{2}\right)^2 = \frac{107}{12}$

$\sqrt{\left(x + \frac{5}{2}\right)^2} = \sqrt{\frac{107}{12}}$

$x = \pm\sqrt{\frac{107}{12}} - \frac{5}{2}$

Solving Quadratic

$x^2 - 8x + 15 = 0$

$x^2 - 8x = -15$

$x^2 - 8x + 16 = -15 + 16$

$(x-4)(x-4) = 1$

$(x-4)^2 = 1$

$x - 4 = \sqrt{1}$

$x = \sqrt{1} + 4$

$4x^2 + x - 3 = 0$

$4x^2 + x - 3 = 0$

$4x^2 + x = 3$

$4x^2 + x + \left(\frac{1}{2}\right)^2 = 3$

$4x^2 + x + \frac{1}{4} = 3$

$\frac{4x^2}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$

$x^2 + \frac{x}{4} =$

Implementing Quadratic Formula

$\frac{ax^2}{a} + \frac{bx}{a} + \frac{c}{a} = \frac{0}{a}$

$x^2 + \frac{bx}{a} + \frac{c}{a} = 0$

$x^2 + \frac{bx}{a} = -\frac{c}{a}$

$x^2 + \frac{bx}{a} + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a}$

$x^2 + \frac{bx}{a} + \frac{b^2}{4a^2} = -\frac{c}{a} + \frac{b^2}{4a^2}$

$x^2 + \frac{bx}{a} + \frac{b^2}{4a^2} = -\frac{c}{a} + \frac{b^2}{4a^2}$

$\left(x + \frac{b}{2a}\right)^2 = \frac{-c \times 4a}{4a} + \frac{b^2}{4a^2}$

$\left(x + \frac{b}{2a}\right)^2 = \frac{(4a)(-c) + b^2}{4a^2}$

$x + \frac{b}{2a} = \sqrt{\frac{(4a)(-c) + b^2}{4a^2}}$

$$x + \frac{b}{2a} = \sqrt{\frac{-4ac + b^2}{4a^2}}$$

$$x = \frac{\sqrt{b^2 - 4ac}}{2a} - \frac{b}{2a}$$

$$x = -b \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

Synthetic division

Used this method to solve the Polynomials which have higher degree

$15x^4 + x^3 - 52x^2 + 20x + 6 = 0$

This is the Expression and we need to solve it by reducing it by binomial Either they Provide the divisior to solve this using the Synthetic Division

Else there will be a trick to Find the Possibilities

Its Not an random trick

let's derive

For example

$x^3 + 2x^2 + 3x + 6 = 0$

$x^3 + 2x^2 + 3x = -6$

$x(x^2 + 2x + 3) = -6$

x must be an fact of '6'

Lets derive the Synthetic Equation

$$15x^4 + x^3 - 52x^2 + 20x + 16 = 0$$

Let's take the Constant '16' and take whole Factors

Factor of '16' = ±1, 1, 2, 4, 8, 16

2	15	1	-52	20	16
	0	30	62	20	80
	15	31	10	40	96

→ Positive '2' Not works

-2	15	1	-52	20	16
	0	-30	+58	-12	-16
	15	-29	6	8	0

→ Negative '-2' is one of the solution

$$(x+2)(15x^3 - 29x^2 + 6x + 8) = 0$$

$$15x^3 - 29x^2 + 6x + 8 = 0$$

Let's do Synthetic division for this equation

Take the Constant and find the Factor

The Factor of '8' is 1, 2, 4, 8, 15 Factor 1, 3, 5

1	15	-29	6	8
	0	15	-14	-8
	15	-14	-8	0

So -1 is one of the solution

$$(x+2)(x-1)(15x^2 - 14x - 8) = 0$$

Quadratic Equation $= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$15x^2 - 14x - 8 = 0$ (a, b, c)

$$= \frac{+14 \pm \sqrt{-14^2 - 4(15)(-8)}}{2(15)}$$

$$= \frac{14 \pm \sqrt{-14^2 - 4(15) - 8}}{2(15)}$$

$$\frac{14 + \sqrt{196 + 480}}{30} \quad \left| \quad = \frac{14 - \sqrt{196 + 480}}{30} \right.$$

$$\frac{14 + \sqrt{676}}{30} = \frac{14 + 26}{30} \quad \left| \quad = \frac{14 - \sqrt{676}}{30} = \frac{14 - 26}{30} \right.$$

$$= \frac{40}{30} = \frac{4}{3} \quad \left| \quad = \frac{12}{30} = \frac{2}{5} \right.$$

Put this $\frac{4}{3}$ and $\frac{2}{5}$ as the answer

$x = \frac{4}{3} \Rightarrow 3x - 4$, $x = \frac{2}{5} \Rightarrow (5x - 2)$

∴ The Overall answer for the Polynomial

$(x+2)(x-2)(3x-4)(5x-2)$ //

This are the Final Answer //

Rational Expression Operation

1) $$\frac{x^2+3x-4}{x^2+4x-21} \div \frac{x^2+x-2}{x^2+9x-14}$$

$$\frac{(x+4)(x-1)}{(x+7)(x-3)} \div \frac{(x+2)(x-1)}{(x+7)(x-2)}$$

$$\frac{(x+4)(x-1)}{(x+7)(x-3)} \times \frac{(x+7)(x-2)}{(x+2)(x-1)}$$

$$\frac{(x+4)(x-2)}{(x-3)(x+2)}$$

2) $$\frac{3x}{x+5} - \frac{6}{x-2}$$

$$\frac{(x-2)(3x) - (x+5)(6)}{(x+5)(x-2)}$$

$$3x^2 - 6x - 6x - 30$$

$$3x^2 - 12x - 30$$

$a \quad b \quad c$

$$(4)(3)(-30)$$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$12 \pm \frac{\sqrt{144+300}}{6} \Rightarrow 12 \pm \frac{\sqrt{500}}{6}$$

22^2

100×5

$$= 12 \pm \frac{\sqrt{100 \times 5}}{6} \Rightarrow 12 \pm \frac{10\sqrt{6}}{6}$$

$$\frac{12 + 10\sqrt{6}}{6}$$

$2 + 10\sqrt{6}$, $2 - 10\sqrt{6}$

$2 + 10\sqrt{6} = x$

$x \cdot 2$

$x = 2 + \sqrt{24}$, $x = 2 - \sqrt{24}$ $-6 \cdot 1$

$x^2 - 4x - 10$

$a \quad b \quad c$

$8 \cdot 8$

$$4 \pm \sqrt{16 - 4 \cdot -10}$$

$16 + 4 \cdot 6$

$$4 \pm \frac{\sqrt{64}}{2}$$

$$\frac{4 \cdot 8}{\sqrt{6}}$$

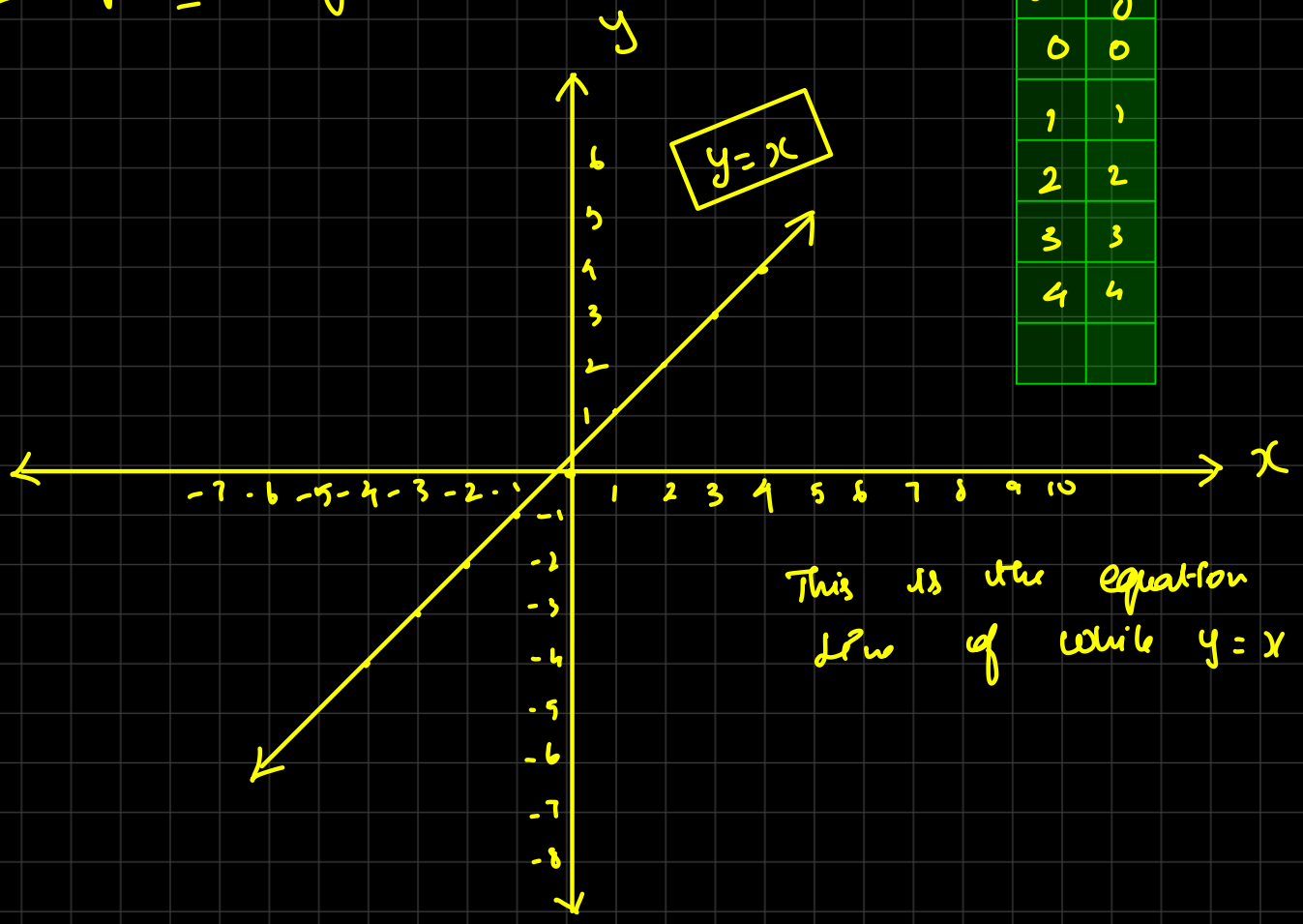
64

8×7

Graphing in algebra

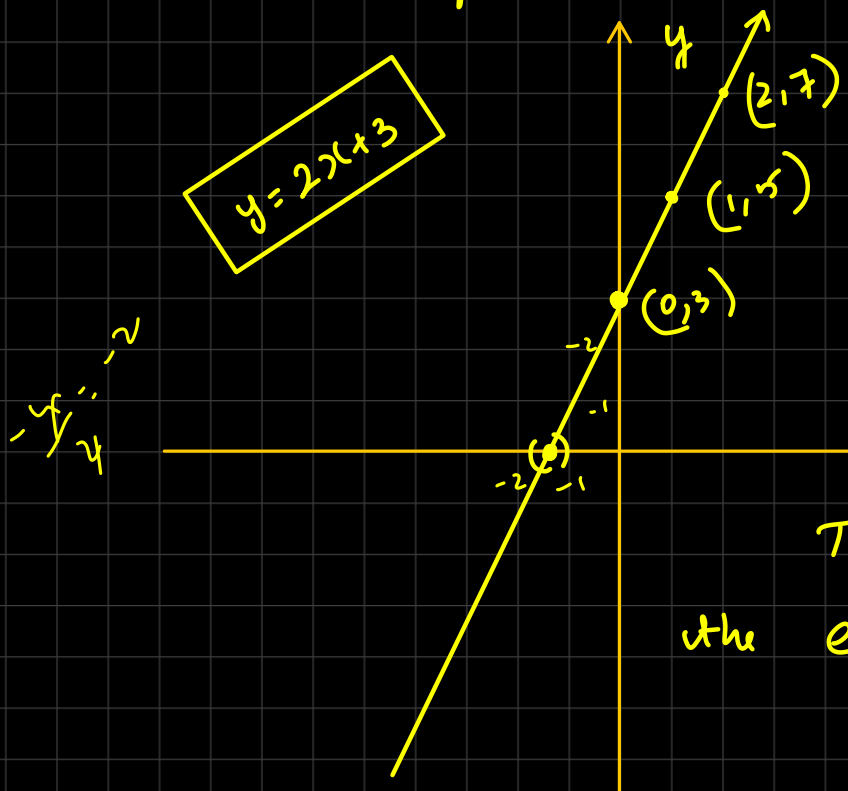
while $y = x$

x	y
0	0
1	1
2	2
3	3
4	4



This is the equation line of while $y = x$

Let's derive the equation while $y = 2x + 3$



$2(0)+3$
$2(1)+3$
$2(2)+3$
$2(3)+3$

x	y
0	3
1	5
2	7
3	9
4	11
5	13

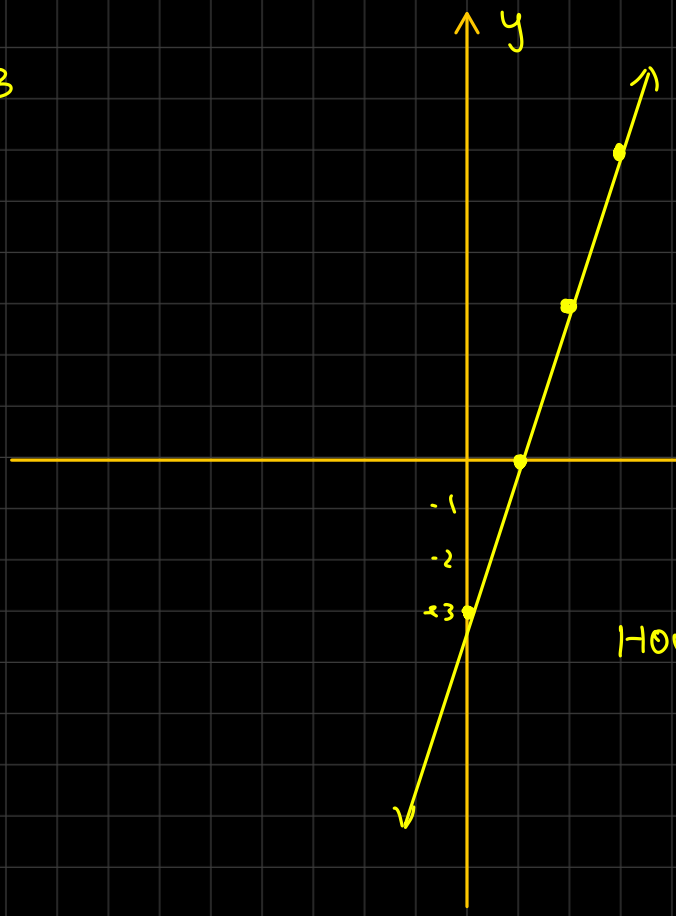
This is the line of the equation How looks like

$$y = 2x + 3$$

Then the line Intercept 'x' In this equation when $y = 0$

$$0 = 2x + 3 \Rightarrow \boxed{x = -\frac{3}{2}}$$

10) $y = 3x - 3$



$3(0) - 3$
$3(1) - 3$
$3(2) - 3$
$3(3) - 3$
$3(4) - 3$

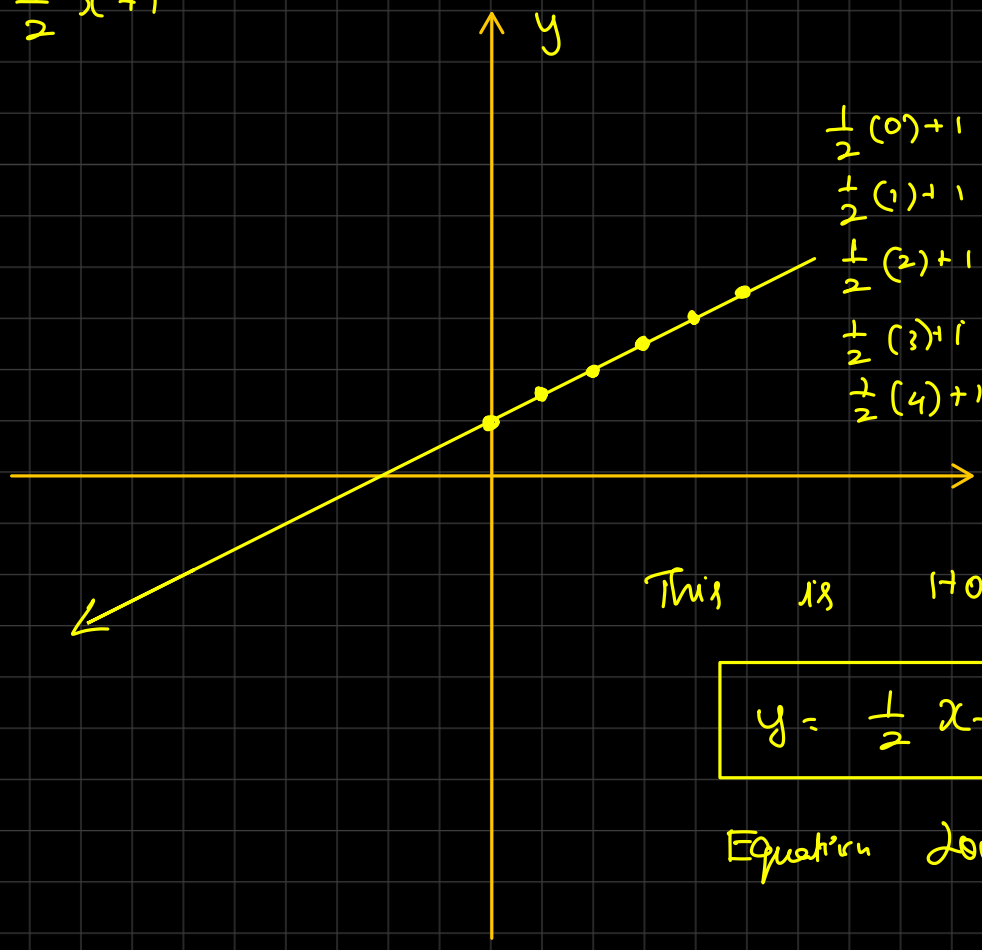
x	y
0	-3
1	0
2	3
3	6
4	9
5	12

This is the line

How the $y = 3x - 3$

Equation look's like

2) $y = \frac{1}{2}x + 1$



$\frac{1}{2}(0) + 1$
$\frac{1}{2}(1) + 1$
$\frac{1}{2}(2) + 1$
$\frac{1}{2}(3) + 1$
$\frac{1}{2}(4) + 1$

x	y
0	1
1	$\frac{3}{2}$
2	2
3	$\frac{5}{2}$
4	3
5	$\frac{7}{2}$

This is How the

$y = \frac{1}{2}x + 1$

Equation looks to

Slope and Intercepter

$$\text{Slope} = \frac{\text{Changes in } y}{\text{Changes in } x}$$

$y = mx + b$

y	Output
m	Slope
x	Input
b	Intercept

How can we find the changes in y we can take any two point of y and see the difference. Same as for x

Example $\overset{x_1\ y_1}{(2,3)}\ \overset{x_2\ y_2}{(5,4)}$

$$\frac{y_2 - y_1}{x_2 - x} \quad \frac{4-3}{5-2} \Rightarrow \frac{1}{3}$$

Intercepter

using the above equation we can find the Intercepter $\overset{x\ y}{(2,3)}$ and slope $= \frac{1}{3}$ $\quad y = mx + b$

$$3 = \frac{1}{3}(2) + b \Rightarrow b = 3 - \frac{2}{3} \quad b = \frac{7}{3}$$

$\overset{x\ y}{(5,4)}$ $\quad 4 = \frac{1}{3}(5) + b \Rightarrow b \Rightarrow 4 - \frac{5}{3} \Rightarrow \frac{12-5}{3} = \frac{7}{3}$

The y-Intercepts $\left(0, \frac{7}{3}\right)$

1) Coordinates $\overset{x_1\ y_1}{(2,0)}\ \overset{x_2\ y_2}{(4,1)}$

$$\text{Slope} = \frac{1-0}{4-2} = \frac{1}{2}$$

Intercepter $= y = mx + b \Rightarrow 0 = \frac{1}{2}2 + b$

2) $\overset{x_1\ y_1}{\left(\frac{3}{2},0\right)}\ \overset{x_2\ y_2}{(1,1)} \Rightarrow \frac{1-0}{1-\frac{3}{2}} = \frac{1}{-\frac{1}{2}} = 1 * \frac{2}{-1} \Rightarrow \frac{2}{-1}$

$\Rightarrow -2$

$= 0 = -2\left(\frac{3}{2}\right) + b \quad \frac{-b}{2}$

$\boxed{b = 3}$

$y = mx + b$

$(x_1, y_1)\ (x_2, y_2)$

$(-6, -3)\ (0, 9) = \frac{9+3}{0+6} = \frac{12}{6} \Rightarrow \boxed{2} \Rightarrow 9 = 2(0) + b \Rightarrow \boxed{b = 9}$

$(x_1, y_1)\ (x_2, y_2)$

$(1, 5)\ (-2, 7)$

$\boxed{y = 2x + 9}$

$= \frac{7-5}{-2-1} = \boxed{\frac{2}{-3}} \Rightarrow 5 = -\frac{2}{3}(1) + b = \frac{17}{3}$

$$\boxed{y = -\frac{2}{3}x + \frac{17}{3}}$$

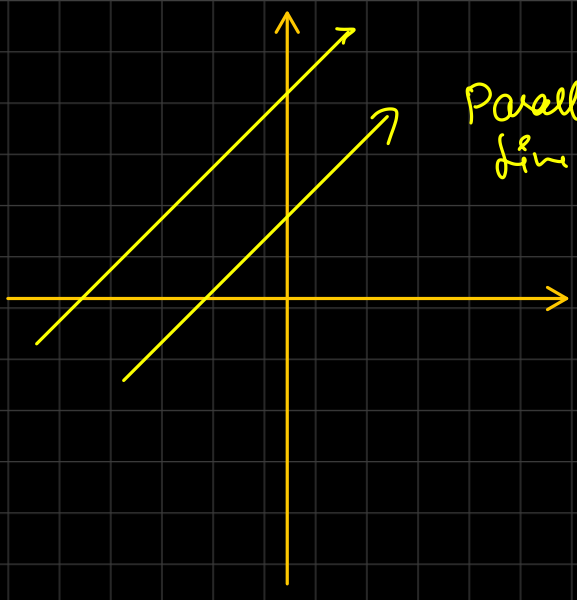
Slope - Intercept Form : $y = mx + b$

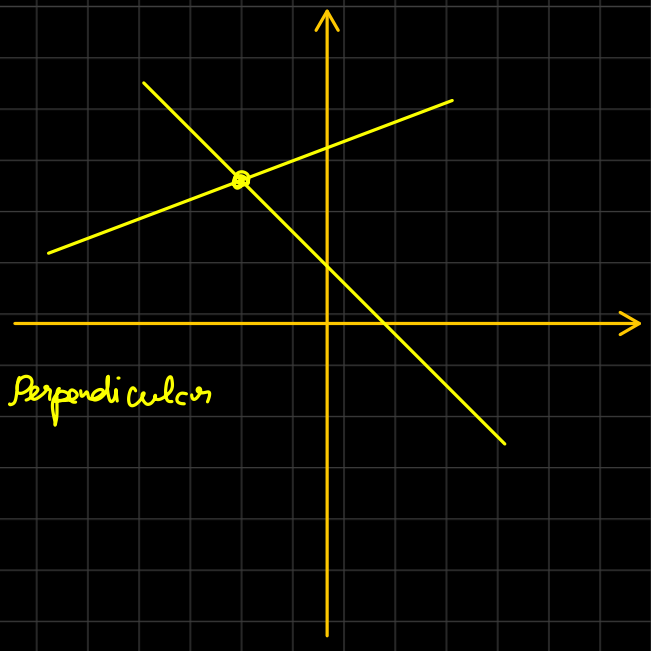
Standard Form = $ax + by = c$

$y = 3x - 2 \Rightarrow -3x + y = -2 \div -$

$$\boxed{3x - y = 2}$$

$y = \frac{2}{3}x + \frac{7}{3}$ $\qquad \frac{2x+7}{3} = y \Rightarrow 3y = 2x + 7 \Rightarrow -2x + 3y = 7$



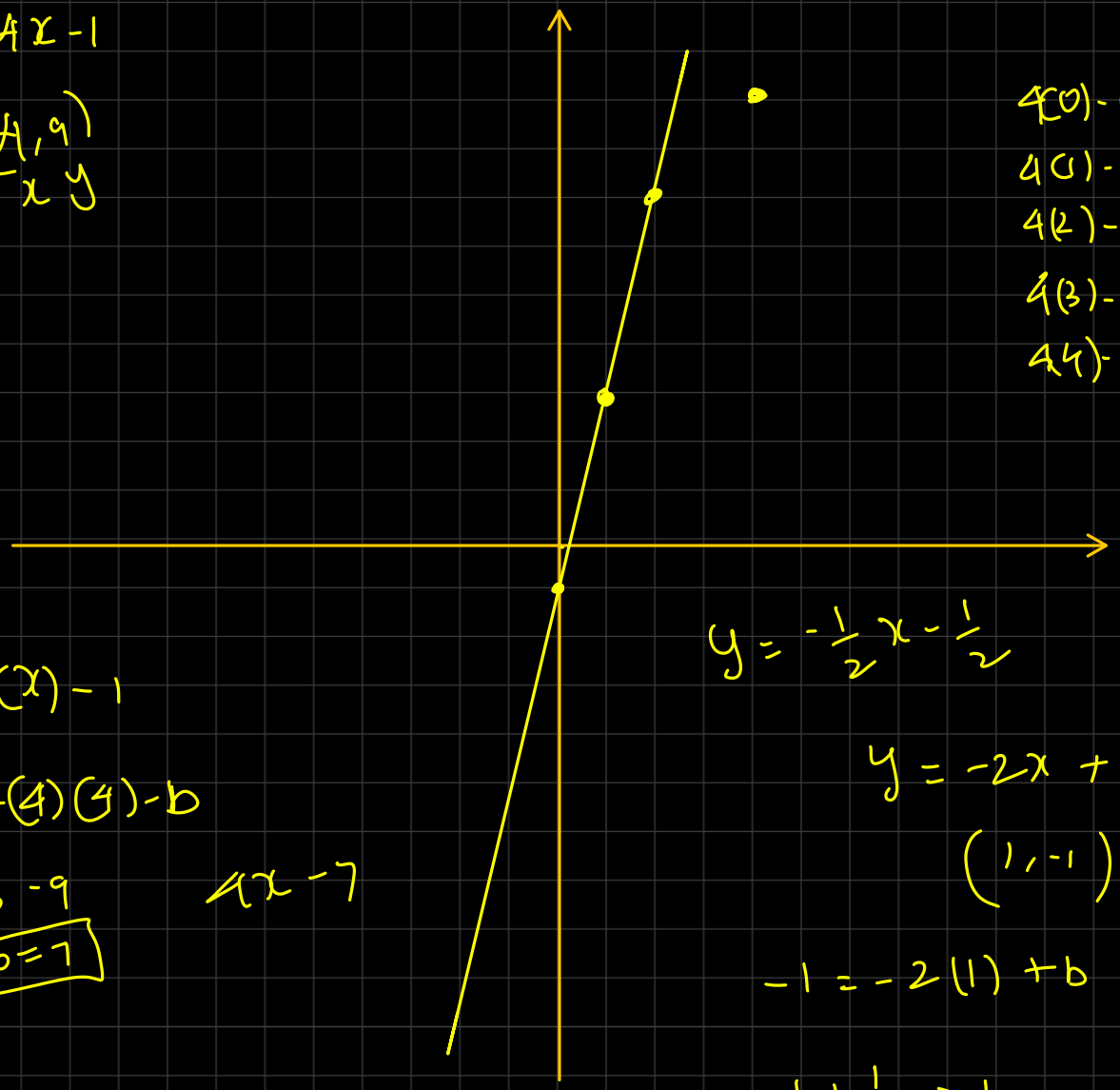


$y = mx + b$

The Old Slope is the Value 'm' is the Other & the Slope is $\frac{1}{m}$

$y = 4x - 1$

$(4, 9)$
$x \; y$



x	y
$4(0)-1$	0 $\quad -1$
$4(1)-1$	1 $\quad 3$
$4(2)-1$	2 $\quad 7$
$4(3)-1$	3 $\quad 11$
$4(4)-1$	4 $\quad 15$

$y = 4(x) - 1$

$9 = (4)(4) - b$

$16 - 9 \qquad 4x - 7$

$\boxed{b = 7}$

$y = -\frac{1}{2}x - \frac{1}{2}$

$y = -2x + 5$

$(1, -1)$

$-1 = -2(1) + b$

$-1 + \frac{1}{2} = b$

$\frac{1}{2} - 1 = b = \boxed{-\frac{1}{2}}$

$y = -2x + b$

$x \quad y$
$(1, -1)$

$-1 = -\frac{1}{2}(1) + b$

$-1 = -\frac{1}{2} + b \Rightarrow \frac{1}{2} - 1 = b$

$\boxed{-\frac{1}{2}}$

Algebra -> two equation solving System with two unknown

There are three methods of solving two Equation

1) Just by looking the Graph

2) Substitution

(i) $10x - 2y = 40$

$10x = 40 + 2y$

$x = 2y + 4$

$4(2y+4) + 2y = 2$

$8y + 16 + 2y = 2$

$10y + 16 = 2 \Rightarrow y = \frac{-14}{10}$

$y = -\frac{7}{5}$

$x = 2y + 4$

$x = 2\left(-\frac{7}{5}\right) + 4$

$x = -\frac{14}{5} + 4 \quad \frac{6}{5}$

$\left(\frac{6}{5}, -\frac{7}{5}\right)$

(ii) Elimination

$5x + 2 = -18$

$5x = -16$

$x = -\frac{16}{5}$

$5x + 2y = -18 \quad \times 5$

$x - 7y = -11 \quad \times 5$

$5x + 2y = -18$

$5x - 35y = -55$

$37y = 37$

$y = 1$

$5x + 2 = -18$

$5x = -16$

$x = -\frac{16}{5}$

$5x + 2y = -18 \quad \times 5$

$x - 7y = -11 \quad \times 25$

$25x + 10y = -90$

$25x + 175y = 275$

$185y = 185$

$y = \frac{185}{185}$

$y = 1$

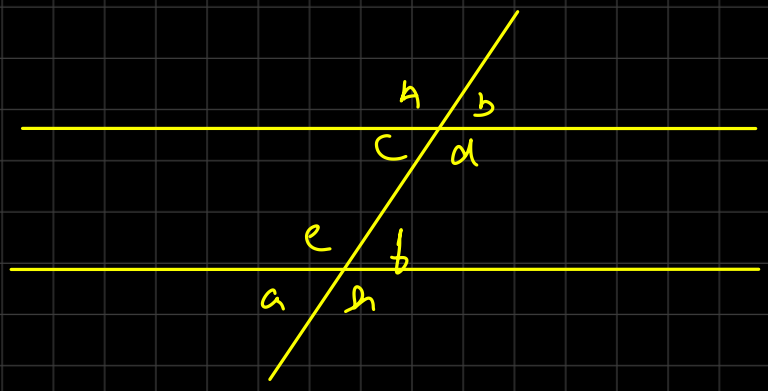
Geometry

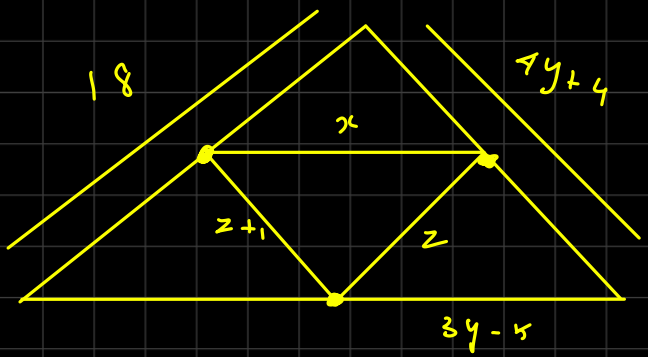
Types of Angle

(i) 90° which is 90° Angle

(ii) 90° ∠ Acute Angle

(iii) 90° ∠ angle < 180 Angle Oblun Angle



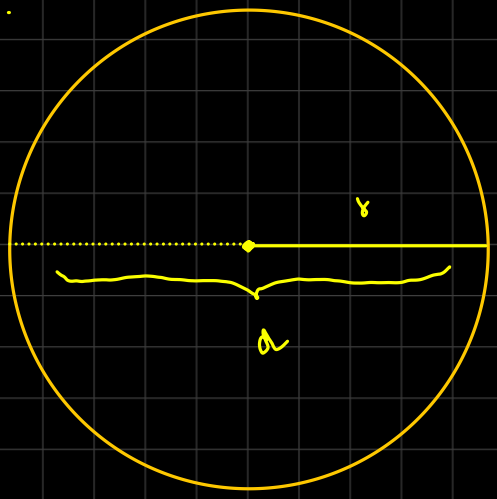


$Z = 9$

$Z+1 = 10$

$4y+4 = 4$

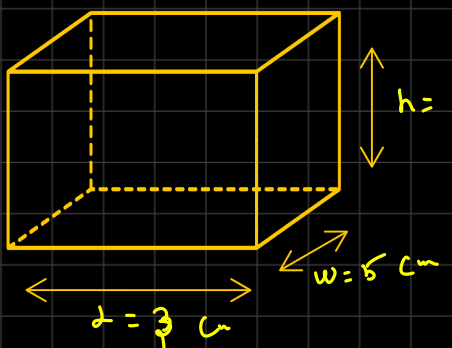
$3y-5 = 7$



Perimeter of Circle

$2\pi r$

Calculating 3d area

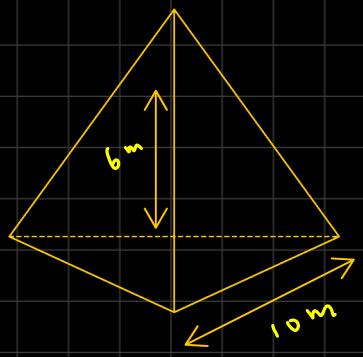


$= 2(l \times h) + 2(w \times h) + 2(l \times w)$

$= 2(3 \times 8) + 2(5 \times 8) + 2(3 \times 5)$

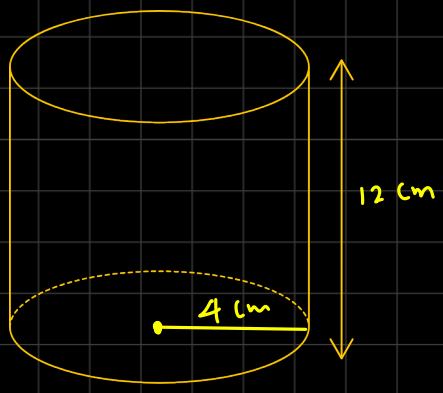
$= 48 + 80 + 30$

$= 158\ cm^2$



= Area of Triangle & square $= 10^2 + 4(\frac{1}{2} \times 10 \times 6)$

$= 100 + 120 = 220\ m^2$



Area of 2 circle + 1 Rectangle

$= 2(\pi r^2) + (2\pi r \times 12)$

$= 32\pi + 8\pi + 12$

$= 128\pi\ cm^2$

Algebra II

Function

Just like our Programming language the function takes input and do calculation and spit out the output.

For example. 30% offer for each item.

So this is how the function derived

$$f(x) = 0.3x$$

$$f(1) = \frac{30}{100} \times 1 \Rightarrow 30.1.$$

$$f(2) = \frac{30}{100} \times \frac{1}{2} \Rightarrow \frac{3}{5}.1. \text{ etc..}$$

This is where function means,

domain → Input
range → Output

We can also solve the algebraic equation to the domain

$$f(x-5) = 2(x) + 10 \Rightarrow 2(x-5) + 10 = 2x - 10 + 10 = 2x$$

We can do all arithmetic operation on two functions.

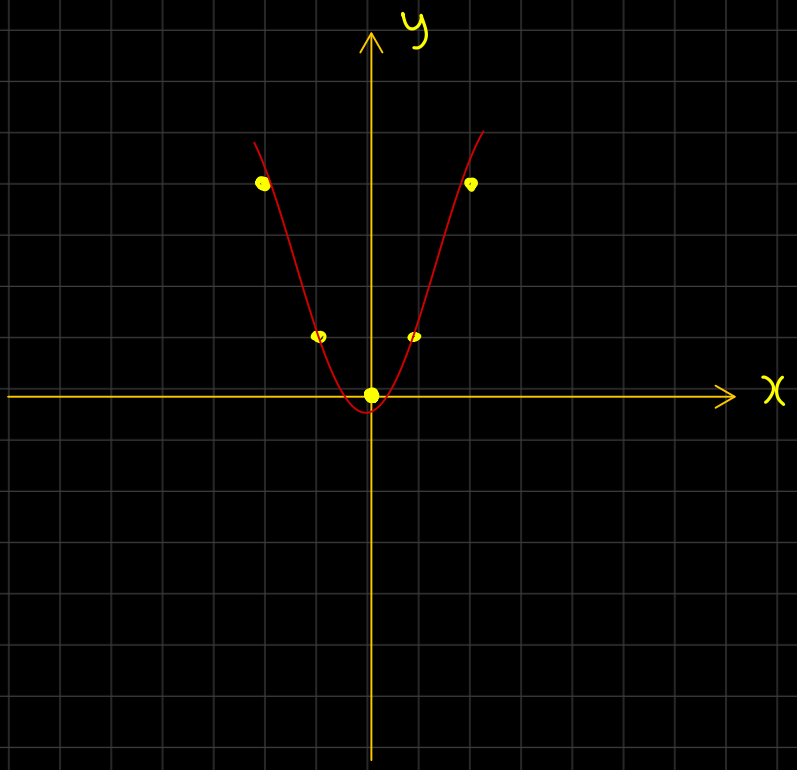
$$f(x) = x + 5$$
$$g(x) = 2x + 4$$

$f(x) + g(x) \Rightarrow (x+5) + (2x+4) = 3x+9$
$f(x) - g(x) \Rightarrow (x+5) - (2x+4) \Rightarrow (x+5) - 2x - 4 \Rightarrow -x - 1$
$f(x) \cdot g(x) \Rightarrow \ldots$
$f(x) \div g(x) \Rightarrow \frac{x+5}{2x+4}$ (Except this we will have real Number as output for all Op)

We can also do nested function called Composite

$$f(g(x)), \quad f(f(x)), \quad f(f(fx))), \ldots$$

Algebric Domain, Range, maxima and minima



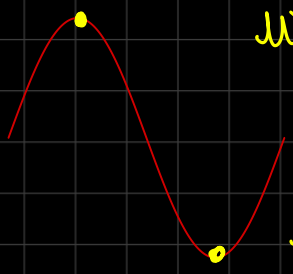
$f(x) = x^2$

x	f(x)
-2	4
-1	1
0	0
1	1
2	4
3	9

$f(x) = x^3$

x	f(x)
-2	-8
-1	-1
0	0
1	1
2	8
3	27

Lets Explain about maxima and minima



where increasing to decreasing Maxima

where decreasing to increasing Minima.

Graphing Algebric Function

Using the function we can Stretch, Compress and reflect the Graph lines

We are just Examinating with Parabola Curve lines

$f(x) = x^2$ Creates the Parabola

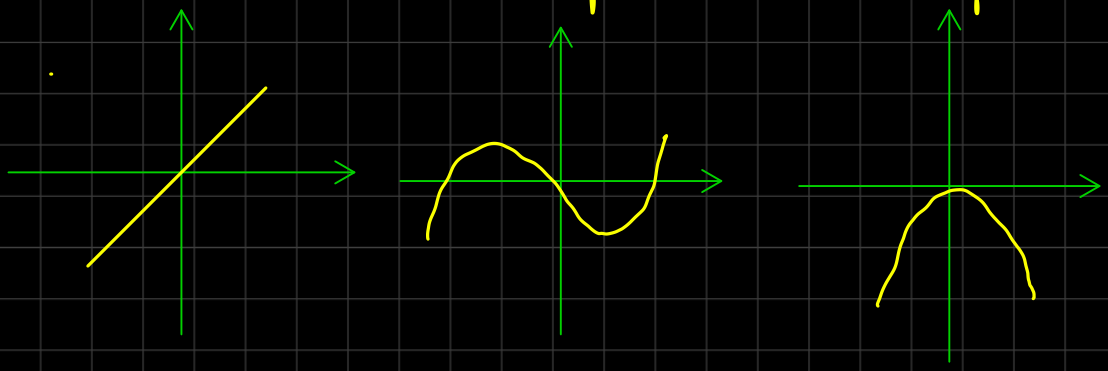
$f(x) = 2x^2$ This Creates the line Stretching

$f(x) = -x^2$ This used to reflects Inverse

$f(x) = -2x^2$ This used to Compress the line.

Continous Function

The Function that does not create Gaps in the Graph line Example:



Like this Not Creating the Gaps in the line

Dis Continous Function

A Function that Creates the Gap in the line for example $\frac{x}{x-1}$ while the x should not be $\boxed{+1}$

So the value become undefined and here How the Graph

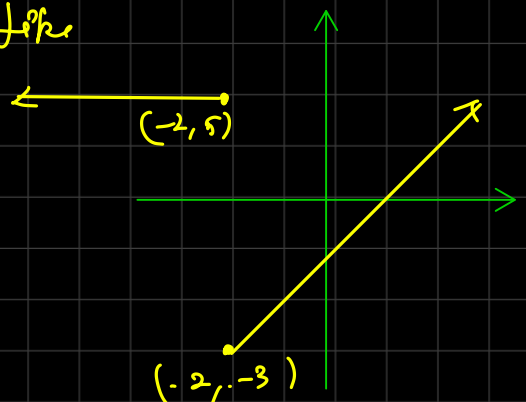
See the Graph Never Touches the Positive (1) So this Creates the Graph in the line that make this dis continous function

Piecewise function

This function have more input and different Output

This is How the function look like

$$f(x) = \begin{cases} 5 & \text{if } x < -2 \\ x-1 & \text{if } x \geq -2 \end{cases}$$



Inverse Function

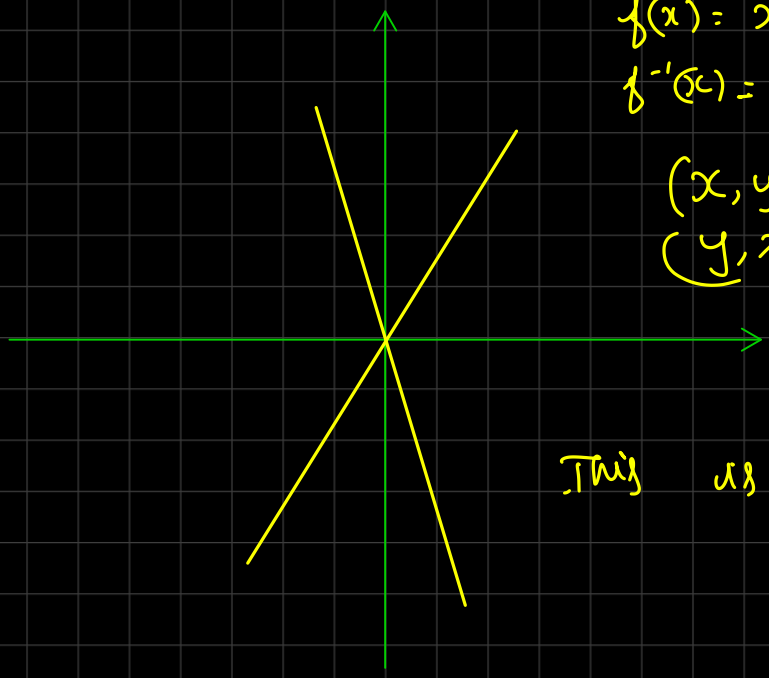
A Function that Actually Get back to it's Original Value of it's 'x' it's Called Inverse Function

Example

$f(x) = x+2$ so if $x = 3$ Then $f(x) = 5$

$f^{-1}(x) = x-2$ so if $x = 5$ Then $f^{-1}(5) = 3$

$f[f^{-1}(x)] = x$ Vise Versa $f^{-1}[f(x)] = x$



$f(x) = x+2$

$f^{-1}(x) = x-2$

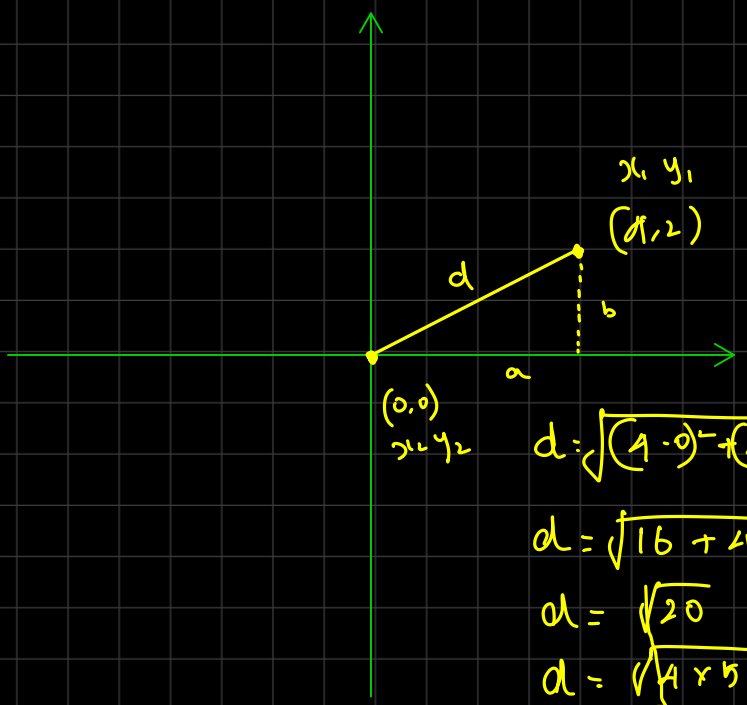
(x, y)

(y, x)

x	f(x)	$f^{-1}(x)$
-2	0	-2
-1	-3	-1
0	2	0
1	3	1
2	4	2
3	5	3

This is How it's look like "Mirror"

Finding the distance between two points



Lets denote 'd' as distance

using Pythagorean Theorem we can find distance

$a^2+b^2=c^2$ $\quad a = x_2 - x_1$ $\quad b = y_1 - y_2$

$d^2 = a^2+b^2$

$d^2 = (x_1-x_2)^2+(y_1-y_2)^2$

$d = \sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$

$d = \sqrt{(4-0)^2+(2-0)^2}$

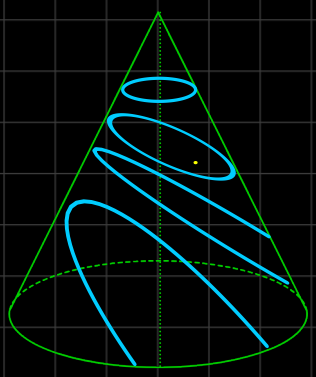
$d = \sqrt{16+4}$

$d = \sqrt{20}$

$d = \sqrt{4 \times 5}$

$d = 2\sqrt{5}$ //

Graphing Conic Section



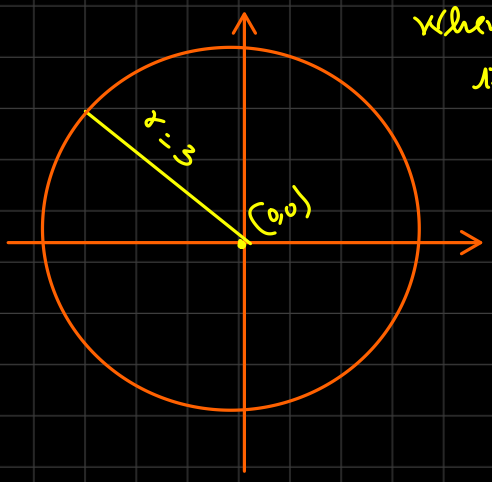
* Circle
* ellipse
* Parabola
* Hyperbola

writing an equation for the circle

Since we know how to find the distance Between two points using Pythagorean theorem using that distance equation we can draw the circle

$$r^2 = \sqrt{(x-h)^2+(y-k)^2}$$

For example if the equation $x^2+y^2=9$



when Origin is (0,0) Since the radius is 3

Since the distance is '3' and we can ask is (3,4) is in the circle

Let's Evaluate

$(x,y) = (3,4)$

$3^2+4^2=3^2 \Rightarrow 9+16=9 \Rightarrow 25 \neq 9$ which is Not

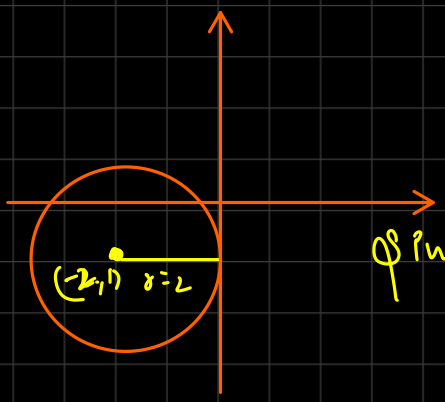
so if the orgin is Not (0,0) let's see

$(x+2)^2+(y+1)^2 = r^2$

$x-h$ $y-k$

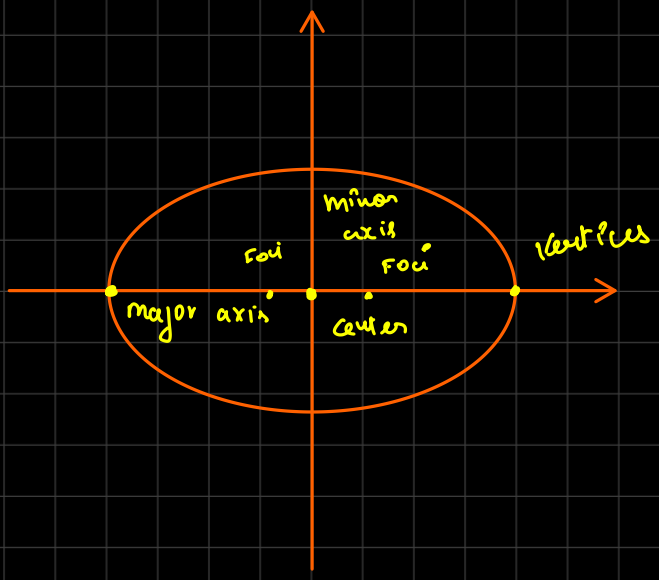
so, the orgin is $(-2,-1)$

This is while the Orgin is Not (0,0)



Since to say $x=-2$ Then radius is '2'

Conic section Part : Ellipse



Foci. $F_1 + F_2$ = Constant

a = major axis

b = minor axis

Vertices = a two end point

Standard Form of ellipse equation :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

when Center $(0,0)$

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

when Origin is (h, K)

Using the Foci and Vertex we Can also Find the a^2 or b^2

since $c^2 = a^2 - b^2$

c : Foci

a : major x

b : minor x

1-) Foci = $(-2,0)$ $(2,0)$

Vertices = $(-4,0)$ $(4,0)$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ Need to find a^2 and b^2

$c^2 = a^2 - b^2$

$2 = 4 - b = \boxed{b = 2}$

$c^2 = 2$

$a^2 = 4$

$$\boxed{\frac{x^2}{4} + \frac{y^2}{2} = 1}$$

2) Center $(-2,4)$

Foci $(-2,3)$ $(-2,5)$

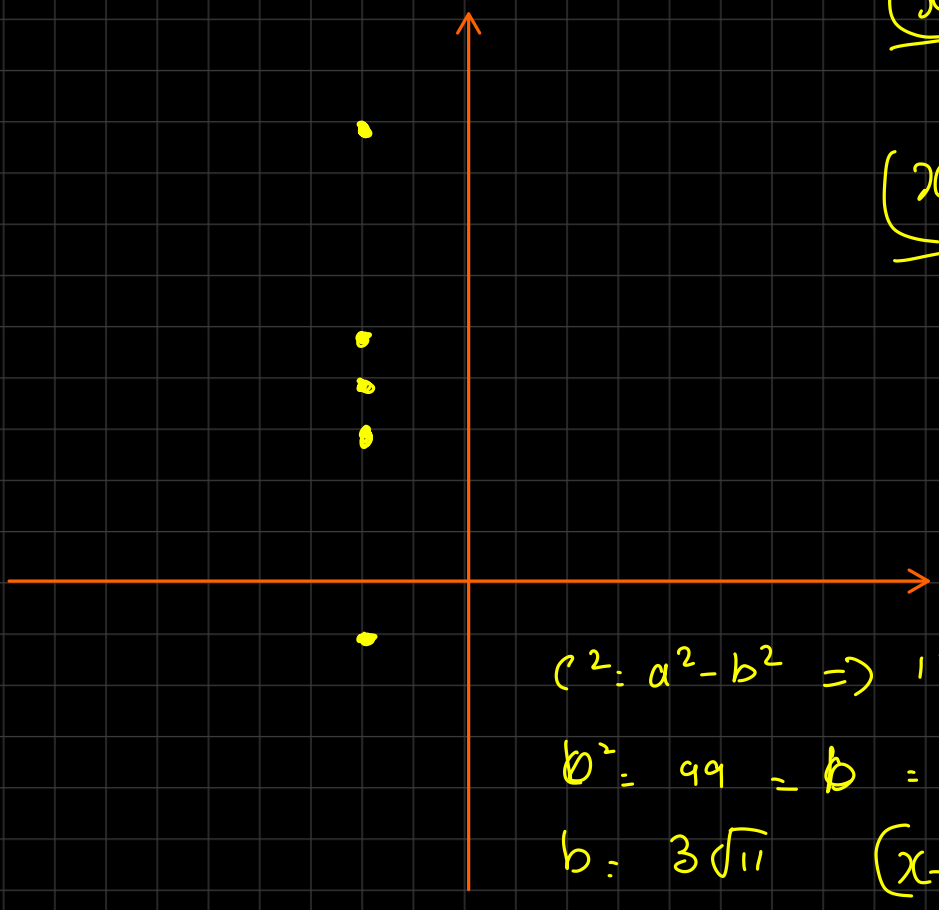
Vertices $(-2,-1)$ $(-2,9)$

$h = -2$

$k = -4$

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x+2)^2}{a^2} + \frac{(y-4)^2}{b^2} = 1$$



$a = 10$

$b = 2$

$c = 1$

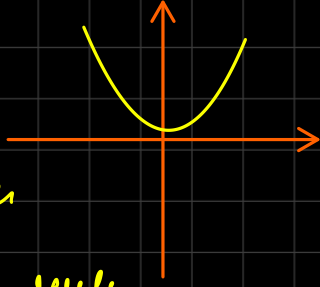
$c^2 = a^2 - b^2 \Rightarrow 1^2 = 10^2 - b^2 =$

$b^2 = 99 \Rightarrow b = \sqrt{99} \Rightarrow b = \sqrt{11 \times 9}$

$b = 3\sqrt{11}$

$$\frac{(x+2)^2}{100} + \frac{(x-4)^2}{3\sqrt{11}} = 1$$

Conic Section - Parabola



We already know what Parabola means that used to form 'U' shape in Graph

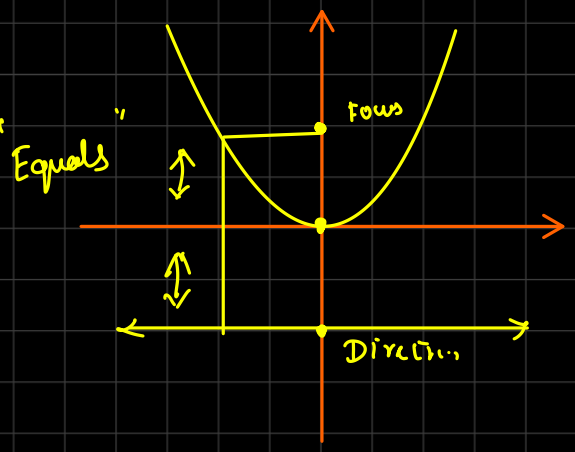
Let's understand the Parabola in Geometric rule

We need two Fixed Point " Focus and directex

Vertex - where the Parabola Curves
Focus - a random point From Vertex to the Curved direction but with same axis Value
directrix - a random point From the Vertex the Opposite direction Value

The Geometric rule is the Distance From the Point to Four and Point to directrix are Equal distance

Distance of the Focus == Distance of the directrix



We said $y = x^2$ and with the Extension of stretch, Shrink and Shift

$$y = a(x-h)^2 + k$$

1a) $2x^2 - 4x + 5$

$2(x^2 - 2x) + 5$

$2(x^2 - 2x + 1 - 1) + 5$

$2(x^2 - 2x + 1) + 2(-1) + 5$

$2(x-1)^2 + 5 - 3$

$$2(x-1)^2 + 3$$

$(-1, 3)$

Vertex

2) $x^2 + 12x + 32$

$x^2 + \frac{12x}{x}6 + 36 - 36$

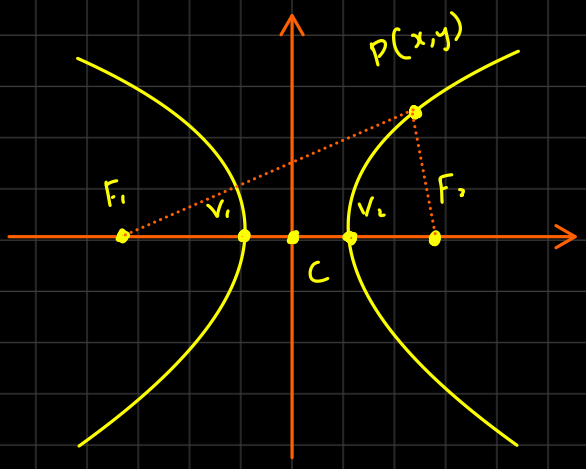
$(x^2 + 12x + 36 - 36) + 32$

$(x+6)^2 - 36 + 32$

$(x^2 + 6)^2 - 4$

$(6, -4)$

Conic section Hyperbola



This is Hyperbola and it's like ellipse but the two foci difference is the constant.

so the formula would be

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

1.) Center at (0,0)
Foci at (−5,0) (+5,0)
Vertices at (−3,0) (+3,0)

a = Center to Vertex

C = Center to Foci

$$\frac{x^2-h^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{3^2} - \frac{y^2}{4^2} = 1$$

$$c^2 = a^2 + b^2$$

$$5^2 = 3^2 + b^2$$

$$b = \sqrt{25-9}$$

$$b = \sqrt{16}$$

$$b = 4$$

Graphing Higher degree Function

Higher degree function has x^3, x^4, x^5 etc...

but How do we sketch them

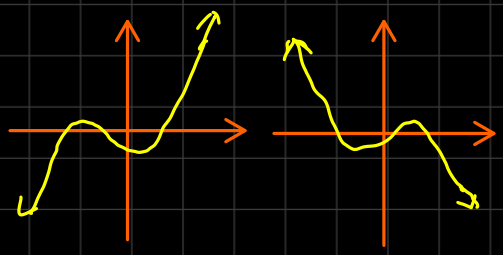
For example $x^3 + 2x^2 + 3x + 1$

see the Higher degree coefficient sign and degree

$(x^3, x^5, x^7 \ldots)$

Odd degree positive (right up, left down)

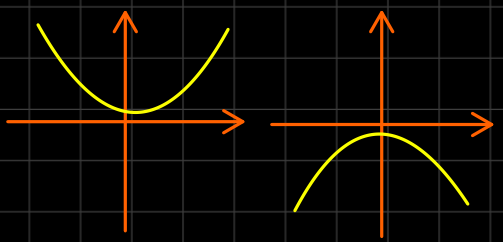
Odd degree Negative left (up, Right down)



$(x^2, x^4, x^6 \ldots)$

even degree positive (both upward)

even degree Negative (both downward)



We Now found the end behaviour

Now we can find the x Intercept For Eample

$$x^3 + 2x^2 + 3x + 1 = 0$$

$$(x+2)(x-1)(x-3)$$

So, The End behaviour is

x^3 which is odd positive

Right upward and left downward

and $x = -2, 1, 3$

So, This is How the Graph would be but we are Not sure about Max and minimes

Multiplicity of Zeros

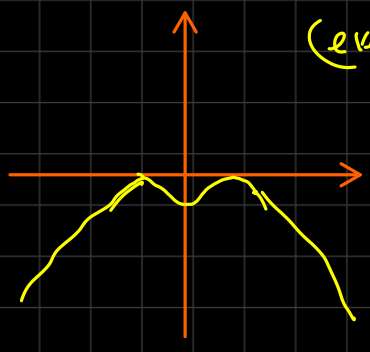
if $x = -2, -2, 1, 3, 3, 3$

If odd multiplicity then Intercepts and Go ∞

If even multiplicity then touches the point and Go to the Opposite direction

(odd)

(even)



Logarithm

Logarithm is an Inverse Operation of Exponent.

$b^x = y$ this is Exponent

$\log_b y = x$ this is Logarithm

Example

$2^3 = 8$

$2^x = 8$ we know its 3 but the right way to express "$\log_2 8 = 3$"

There is some Criteria of the logarithm

$b > 0$, $b \neq 1$ => because some Negative base does not Give real value for Example $-2^{\frac{1}{2}}$ there is no real value and base '1' always Gives '1'

$x > 0$, Example $2^y = x$ => $2^2 = 4$, $2^1 = 2$, $2^0 = 1$, $2^{-1} = \frac{1}{2}$, $2^{-2} = \frac{1}{4}$ see none of them Produces 0 or -n.

Types of Logarithm

Common logarithm => log of No base denoted as 10

Natural Logarithm => denoted as ln means $\log_e x$ => $\ln x$

what 'e' means $e = 2.1783\ldots$ a special value just like a 'π'. It's a Continious Growing Number "'e' is just a constant number of continous Growth".

Change of Base Property => $\log_b x = \frac{\log_a x}{\log_a b}$

Expression and Properties of logorithm

Product rule = $\log(ab) = \log a + \log b$

Quotient rule = $\log(a/b) = \log a - \log b$

power rule = $\log x^a = a \log x$

Comprehension

1.) $\log_3(81x^3) \Rightarrow \log_3 81 + \log_3 x^3 \Rightarrow \log_3 9^2 + \log_3 x^3$

$\Rightarrow 2\log_3 9 + 3\log_3 x \Rightarrow 2 \times 2 + 3\log_3 x = 4 + 3\log_3 x$

2) $\ln(x^2 y^3 / z^4) \Rightarrow \ln x^2 + \ln y^3 - (\ln z^4) \Rightarrow$

$2\ln x + 3\ln y - 4\ln z$

3) $2 + 2\log_5 x \Rightarrow 2 + \log_5 x^2 \Rightarrow \log_5 25 + \log_5 x^2$

$\Rightarrow \log_5(25x^2)$

4) $\frac{1}{2}\left[(\log_2 x^6) - (\log_2 x^4)\right] \Rightarrow$

$= \frac{1}{2}\left[(\log_2 x^6) - (\log_2 x^4)\right]$

$= \frac{1}{2}\log_2(x^6/x^4)$

$= \frac{1}{2}\log_2 x^2$

$= \log_2 (x^2)^{\frac{1}{2}} = \log_2 x$

$(3+2i)+(5-i) = 8+2i-i = 8+i$

$(6-7i)-(4-11i) = 6-7i-4+11i = 2+4i$

$(9-i)(2+2i) = 18-2i+18i-2i^2$

$= 18-16i-2-1 \quad 20+16i$

$(3+i)/(4-i) \quad \frac{(3+i)}{4-i} \times \frac{(4+i)}{4+i} = \frac{12+4i+3i+i^2}{(4-i)^2}$

$= \frac{12+7i+i^2}{(4-i^2)} = \frac{11+7i}{(4-i)^2} = \frac{11+7i}{16+1} = \frac{11+7i}{17}$

$\frac{11}{17} + \frac{7i}{17}$

Set theory

"A set is a collection of Object"

Inclusive set denoted in Square bracket "[]"

[1, 3] which mean 1 and 3 also included

Non inclusive set denoted in the parantheses "()"

(1,3) 1 and 3 Not included

Infinite set $(-\infty, \infty)$

and we can write combination of inclusive and Non inclusive (3, 5] No 3 included but 5

A set can be represent in curly braces Also {}

Union -> combines all distinct values

$A = \{1,2,3\}$ $b = \{1,5\}$ $A \cup B = \{1,2,3,5\}$

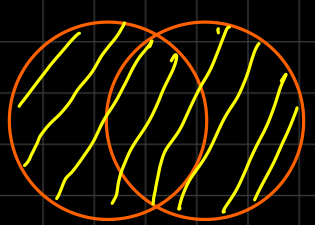
Intersection -> Common Items

$A = \{1,2,3\}$ $b = \{1,5\}$ $A \cap B = \{1\}$

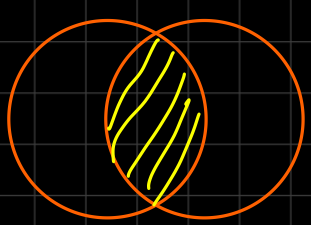
It can be also empty {} or $\emptyset$ null set

Ven Diagram

$A \cup B$



$A \cap B$



Sequence

$\{a_n\} = n$

Factorial

Factorial represent in Exclamation Symbol '!' and we derive in this formal $3! = 3 \times 2 \times 1$

Summation Notation

The Summation Notation represent in the Symbol of Sigma Σ

$$\sum_{\text{start}}^{\text{End (n times)}} \text{Expression} \Rightarrow \sum_{n=1}^{n=3} n^2 = \{1^2, 2^2, 3^2\}$$

Probability

The Probability is a Possibilities of the Outcome. For example if we toss the coin and we ask what is the posibilites of Head Coins have two faces head and tails so in 2 there is one Possibilities so the formula is

$$\text{Possibilities} = \frac{\text{Favourable Outcome}}{\text{Total No of Possibilities}}$$

If three red ball and two blue ball and the possibility of Picking red ball is $\frac{3}{5}$ 60%

Permutation

The Permutation is finding possibilites where the Order matters.

ABCD we Need the possibilities of Picking two Position AB, BA, AC, CA, AD, DA, BC, BD, CB, CD, DB, DC

Totally 12 Outcomes

In the first Position there are 4 Possibilities and for the Second Position there are 3.

so $4 \times 3 = 12$

Permutation formula $_nP^r = \frac{n!}{(n-r)!}$

n = No. of Poss
r = rate of Obj

Combination

Combination is the possibilities where the Order doesn't matter

For Example ABCD. We Can randomly Pick two Peoples so AB Can also BA AB, AC, AD, BC, BD, CD, so '6' outcomes same as that but removed duplicates

Combination Formula $_nC_r = \frac{n!}{r!(n-r)!}$

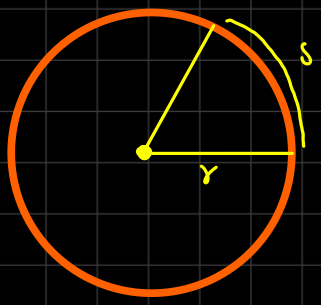
The extra r! is to remove duplicates

Trigonometry.

We have a degree to measure the angle but using the natural circle rythm we can also measure the angle with the help of radian

Radian

What radian actually the angle that we to measure with the help of the radius



when we have the same distance of the arc as radius which is one radian

$$\text{radian} = \frac{\text{Distance of Arc}}{\text{radius}} = \frac{r}{r} = 1$$

$2\pi r$ is the Circumference which is 360°

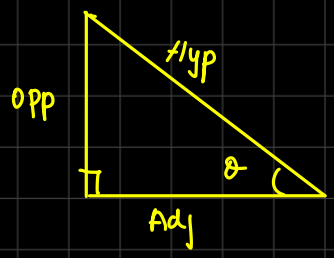
$\frac{2\pi r}{r} = 2\pi$ radian which is full Circle

$\frac{2\pi r}{2} \quad \pi r = \frac{\pi r}{r} = \pi$ radian which is 180° ...

Converting Degree to radian $= \frac{D}{180} \times \pi$

Converting Radian to Degree $= \frac{R}{\pi} \times 180^\circ$

Trigonometry Functions



We Call the angle as 'θ'

Human Can measure the Thing just with the help of Angle and distance.

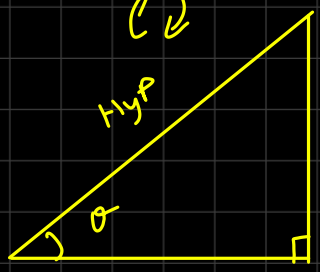
"SOH CAH TOA"

Reciprocal

$\sin\theta = \frac{Opp}{Hy}$ $\quad$ $\operatorname{Cosec}\theta = \frac{Hy}{Op}$

$\cos\theta = \frac{Adj}{Hy}$ $\quad$ $\sec\theta = \frac{Hy}{Adj}$

$\tan\theta = \frac{Op}{Adj}$ $\quad$ $\cot\theta = \frac{Adj}{Op}$



This function just used to Find the ratio's of the two sides

One unit Circle

The One unit Circle is the radius of One circle (r = 1)

In the unit Circle (x,y) is (Cos θ, Sin θ) as Coordinates

so in unit Circle the radius is always 1

and $\cos\theta = \frac{Adj}{Hy} = \frac{x}{1} = x$, $\sin\theta = \frac{Opp}{Hy} = \frac{y}{1} = y$

$(x,y) = (\cos\theta, \sin\theta)$

If you found the x, y in the Unit Circle its simply the x, y is the angle of ratio.

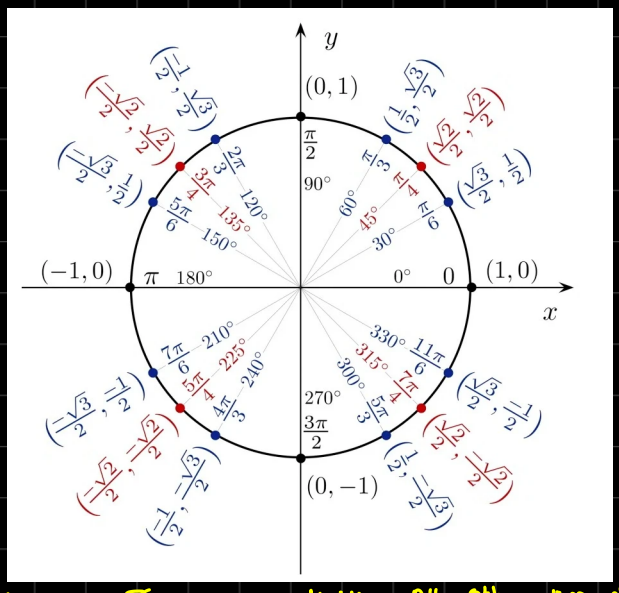
How the values of x and y Calculated

Taking square 1 cm.

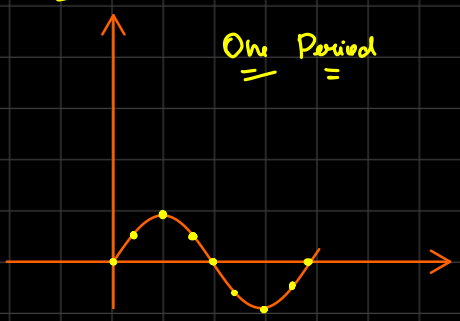
. Need to Find Cos 45° Using Pythogorous

$1^2 + 1^2 = c^2 = 2 = c^2 \Rightarrow c^2 = \sqrt{2}$. Making the Hyp line to '1' by dividing $\frac{\sqrt{2}}{\sqrt{2}} = 1$ and dividing All other two sides

$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ $\cos 45° = \frac{1}{\sqrt{2}}$ Negating $\sqrt{2}$ in denominator $\frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{(\sqrt{2})^2} = \frac{\sqrt{2}}{2}$ $\boxed{\cos 45° = \frac{\sqrt{2}}{2}}$



Let's Graph Trigonometric Function



One Period

$y = \sin x$

$0, 0$

$\frac{\pi}{6}, \frac{1}{2}$

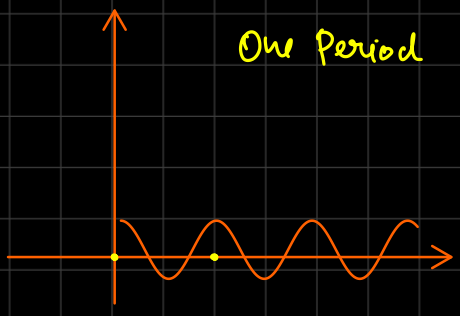
$\frac{\pi}{2}, 1$

$y = a \sin x \Rightarrow$ V stretch (amplitude)

$y = \sin bx \Rightarrow$ H stretch

$y = (\sin x) + K \Rightarrow$ V shift

$y = \sin (x + K) \Rightarrow$ H shift



One Period

$y = \cos x$

$0, 1$

$\frac{\pi}{6}, \frac{\sqrt{3}}{2}$

$y = a \cos x \Rightarrow$ V stretch

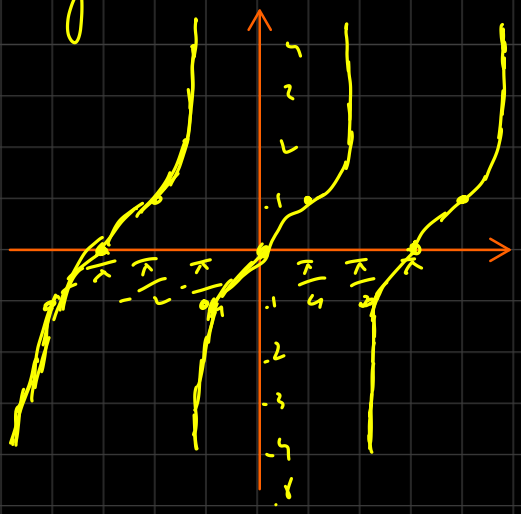
$y = \cos bx \Rightarrow$ H stretch

$y = (\cos x) + K \Rightarrow$ V shift

$y = \cos (x + K) \Rightarrow$ H shift

btw they both look similar

Tangent Two Period One Period Two Period



$\tan \theta = \frac{opp}{adj} = \frac{\sin \theta}{\cos \theta}$

It tell the slop ratio

Inverse Trigonometry Function

$y = \sin x \Rightleftharpoons y = \sin^{-1} x$

$\sin$ value ↔ angle

angle ↔ $\sin$ value

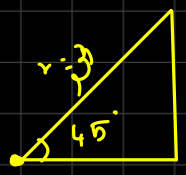
Using the (ratio) value we can Finding Angle

It's Also Called Arc $\sin x$

Polar Coordinates

We just use the x, y as the coordinates to draw a Graph in the rectangular and Polar Coordinates is used to (r, θ)

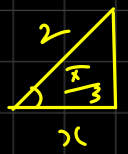
r = radius, θ = Angle



$r = 3, \quad \theta = 45°$

We can able to Convert Polar to rectangular

$(2, \pi/3)$



$\sin\frac{\pi}{3} = \frac{y}{2} = \sin\frac{\pi}{3} \times 2 =$

$\frac{\sqrt{3}}{2} \times 2 = \sqrt{3}$

$\cos\frac{\pi}{3} = \frac{x}{2} = x = 1$

$y = \sqrt{3}$

$x = 1$

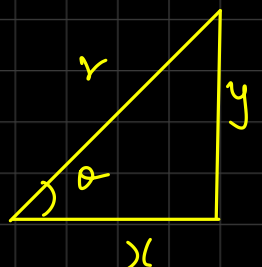
$\left(2, \frac{\pi}{3}\right) \Rightarrow (1, \sqrt{3})$ Converted successfully.

We can Convert the (x, y) equation to (r, θ)

$x + y = 3$

$\cos\theta = \frac{x}{r} = x = r\cos\theta$

$\sin\theta = \frac{y}{r} = y = r\sin\theta$



$x + y = 3 \Rightarrow r\cos\theta + r\sin\theta = 3 \Rightarrow r$ comm

$r(\cos\theta + \sin\theta) = 3 \Rightarrow r = \frac{3}{\cos\theta + \sin\theta}$

Calculus

My understand about the Calculus is we can't able to Calculate everything using our basic mathematic stuff like algebra, Trignometry for example the car is moving the average speed of 40 Km/hr From 11:00 AM to 12:00PM to the distance of 100 km. do we able to know what is the exact speed at 11:30 so where the Calculus invented the thing that often Changes. that is Called "Differential" an the thing that able to Calculate the area under the curve is "Integration".

"As much as we Get the Infinite Solution you Get finet answer"

No maths Get that an Instant Value so the Calculus Comes in

Differential Calculus - Instant Changing Calculus

Integral Calculus - Accumulation. using tiny Infinite Amount to Calculate Finet answer